Capture plan, second real data set

24 layouts x 4 rounds = 96 scenes (5 readings each). One page per layout, in capture order.

Arena centre (the + on every page): Motive x = -0.28 m, y = -0.58 m. Mark it on the floor with tape.

Box positions are given from that centre in cm (dx along Motive x, dy along Motive y) and in Motive metres.

Fastest way: place the boxes roughly, then run python -m tools.place_check --layout Rxx : it reads OptiTrack and

says, live, how far every box (and, for the random family, every robot) is from its target. Within 5 cm is fine.

Triangles: box orientation, by eye. Blue circles: robots with heading. Red hatched: obstacle (about 30 cm).

Yellow: white object that is not a box. Dotted blue circle: the circle the robots stand on.

Days

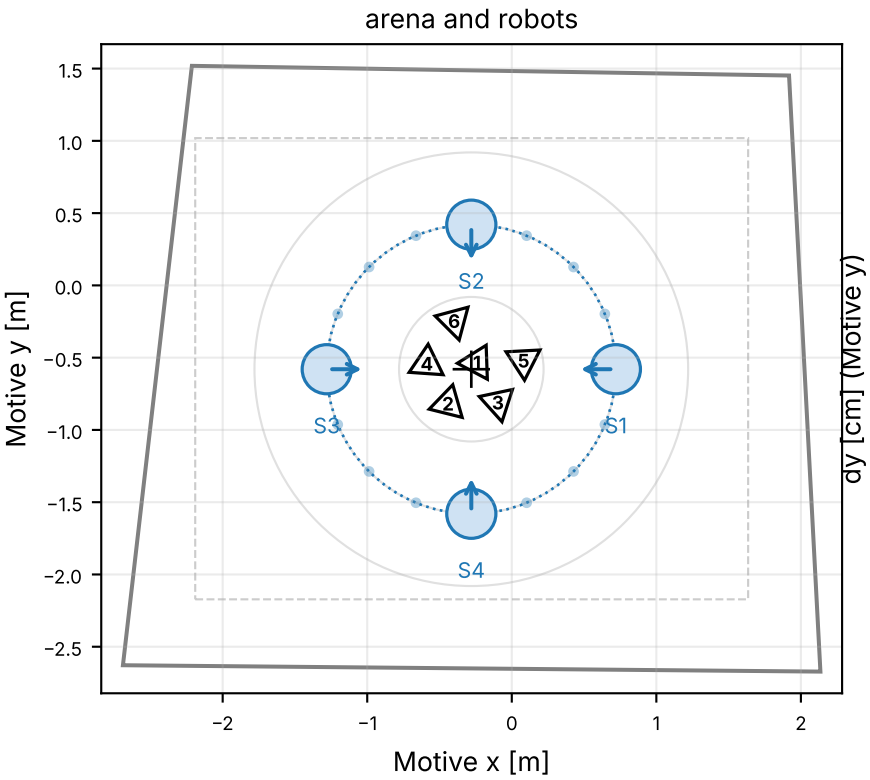
day 6: R81-R104 (24 layouts; boxes 3, 4, 5, 6, 7)

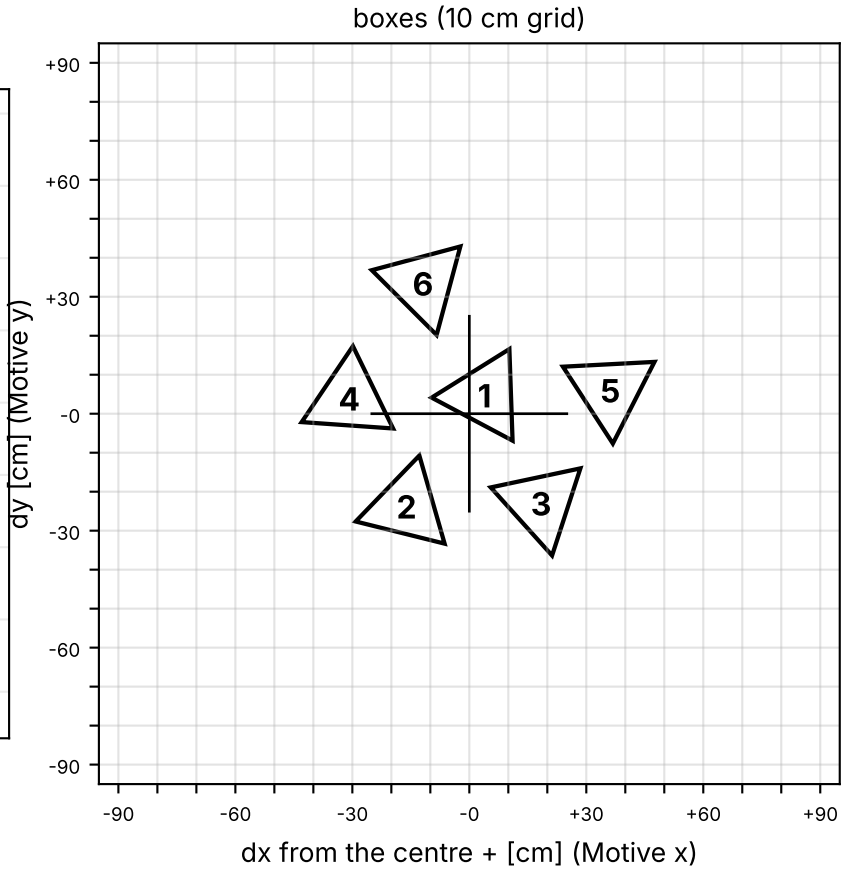
Rules

- Once, before starting: find which way Motive +x and +y point on the floor (slide a box and watch place_check) and mark them with tape arrows at the centre.
- After the first 2-3 layouts of day 1: commit and push, and wait for the check before going on.
- place_check: Enter starts the capture of that layout (launch_all with the right options).
- --box lists only the boxes on the floor (ids 11-17 in this order); the others out of the arena.
- Keep --camera-offset-forward-m 0.12 (the pipeline corrects the lens offset).
- Obstacle layouts: the robots drive along the circle; keep an eye on them near the obstacles.
- End of each day: git add data/survey_v2 docs/capture_plan_v2.csv, commit, push.

R81 day 6 M medium (test)

6 boxes, dense block, not touching, robots: cerchio 1.0 | medium in 4/4 rounds (77% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+4	+5
2	-16	-24
3	+18	-23
4	-31	+4
5	+36	+6
6	-12	+33

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

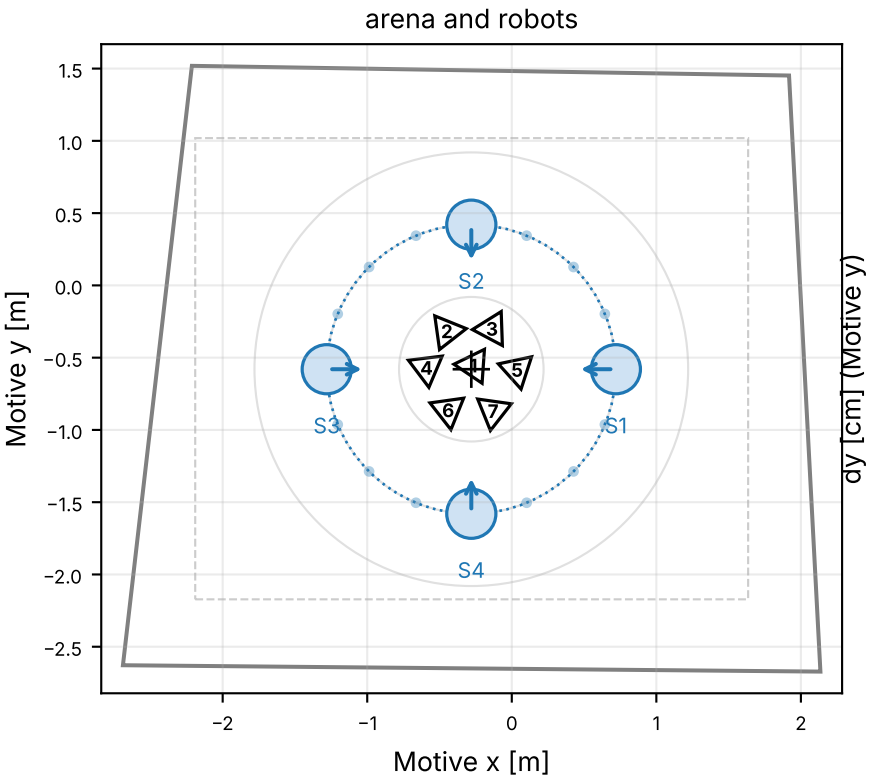
Place and capture: python -m tools.place_check --layout R81

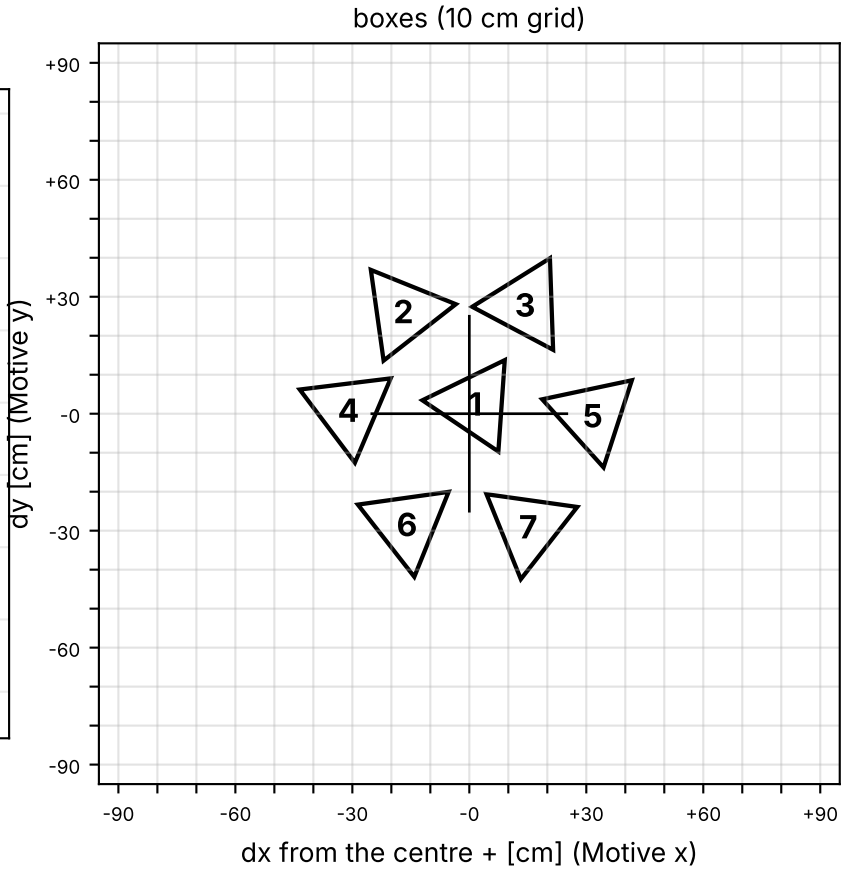
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R81 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R82 day 6 M medium (test)

7 boxes, dense block, not touching, robots: cerchio 1.0 | medium in 3/4 rounds (62% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+2	+2
2	-17	+26
3	+14	+28
4	-31	+1
5	+32	-1
6	-16	-28
7	+15	-29

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

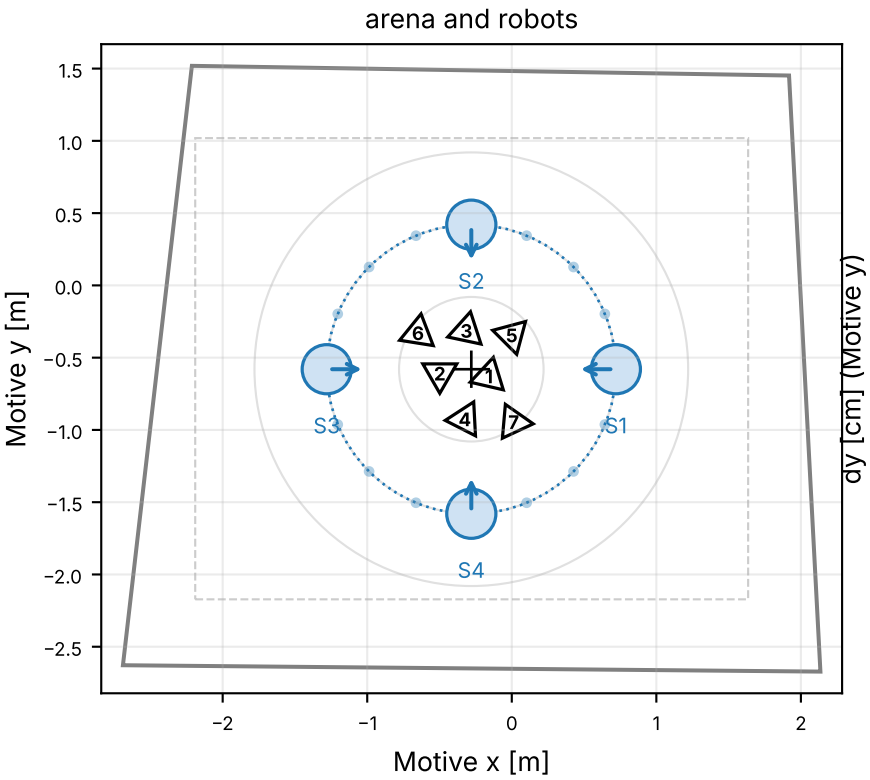
Place and capture: python -m tools.place_check --layout R82

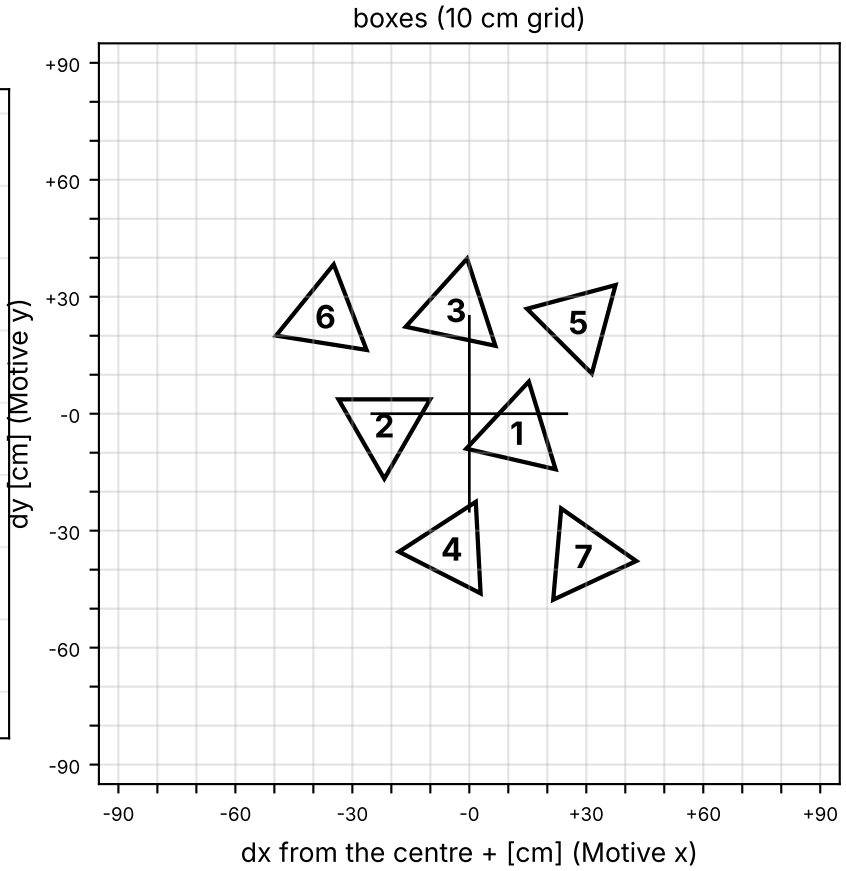
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R82 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R83 day 6 M medium (test)

7 boxes, dense block, not touching, robots: cerchio 1.0 | medium in 4/4 rounds (77% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+12	-5
2	-22	-3
3	-3	+26
4	-4	-35
5	+28	+23
6	-37	+25
7	+29	-37

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

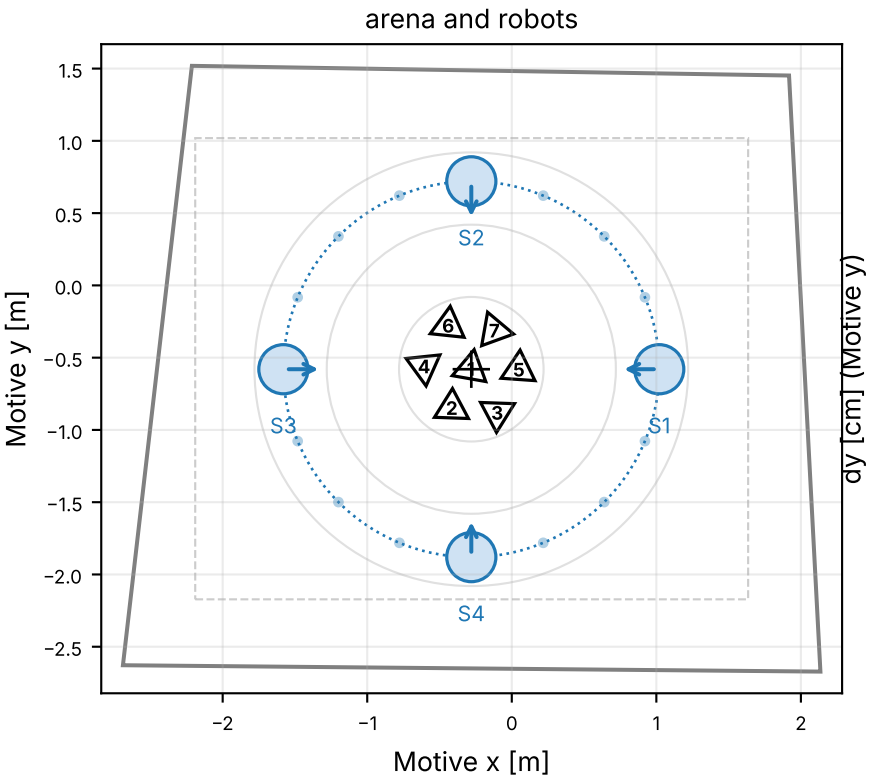
Place and capture: python -m tools.place_check --layout R83

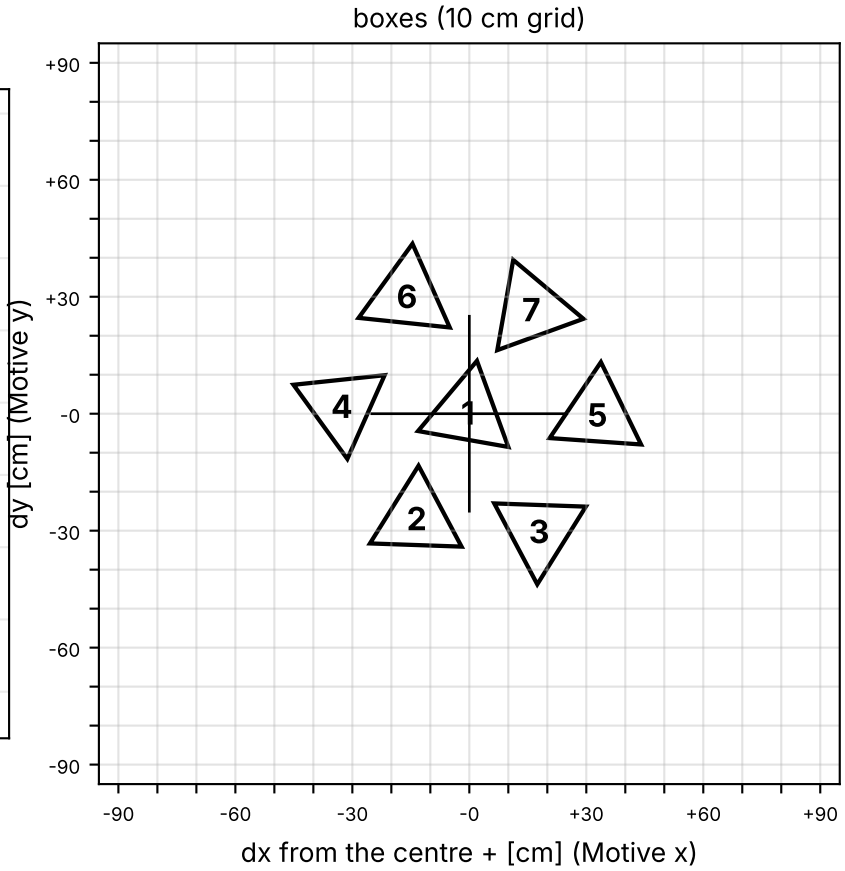
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R83 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R84 day 6 M medium (test)

7 boxes, dense block, not touching, robots: cerchio 1.3 | medium in 4/4 rounds (79% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-0	+0
2	-13	-27
3	+18	-30
4	-33	+2
5	+33	-0
6	-16	+30
7	+16	+27

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

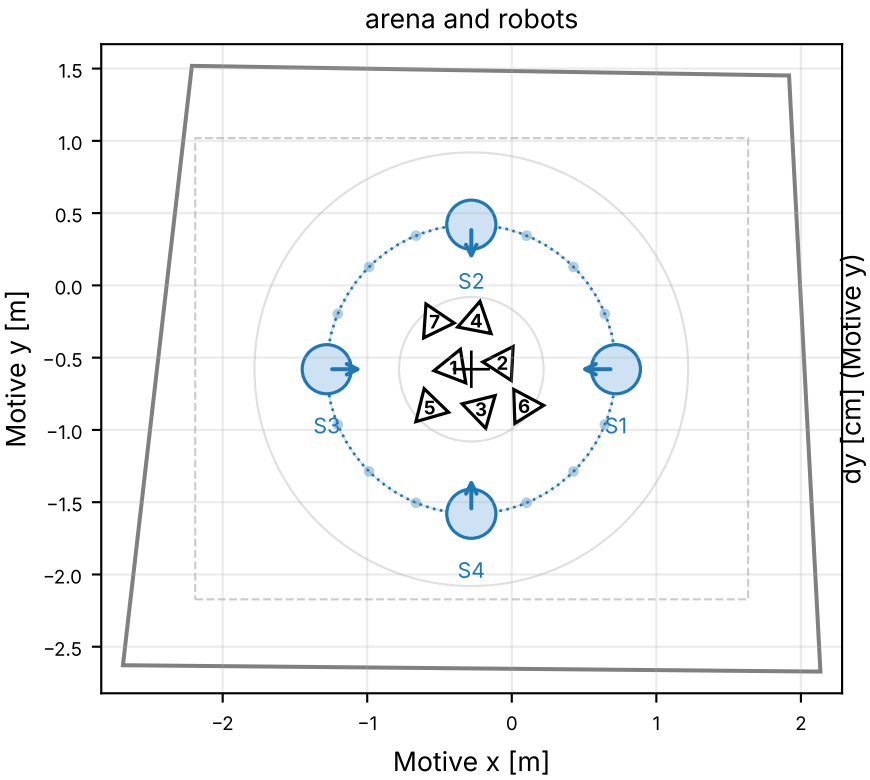
Place and capture: python -m tools.place_check --layout R84

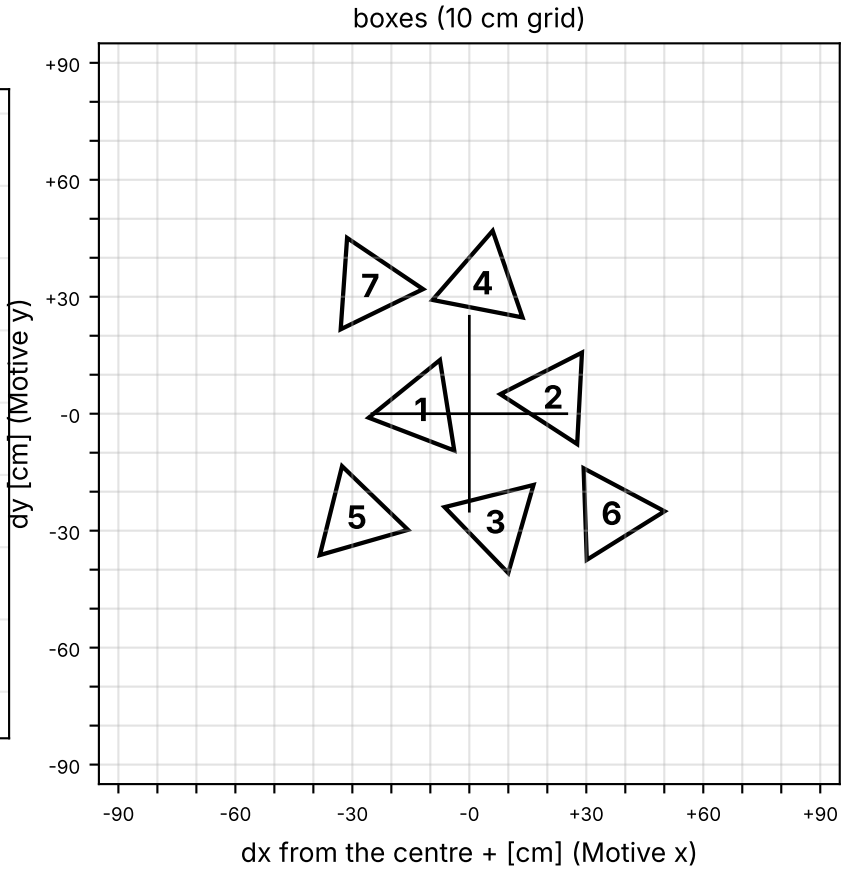
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R84 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R85 day 6 M medium (test)

7 boxes, dense block, not touching, robots: cerchio 1.0 | medium in 4/4 rounds (73% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-12	+1
2	+22	+4
3	+7	-28
4	+3	+34
5	-29	-26
6	+37	-26
7	-25	+33

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

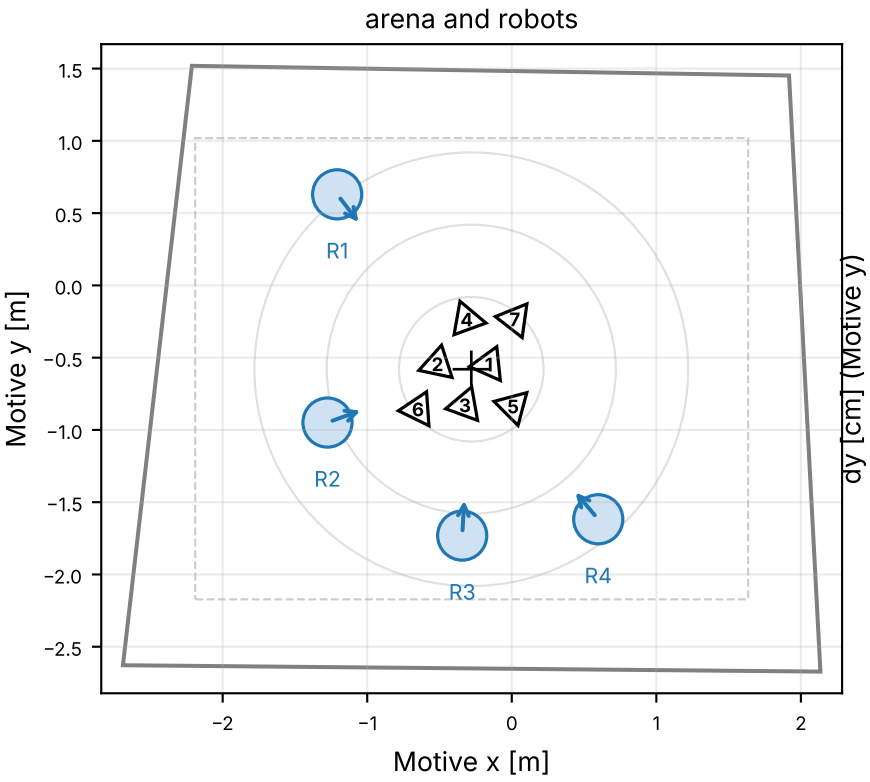
Place and capture: python -m tools.place_check --layout R85

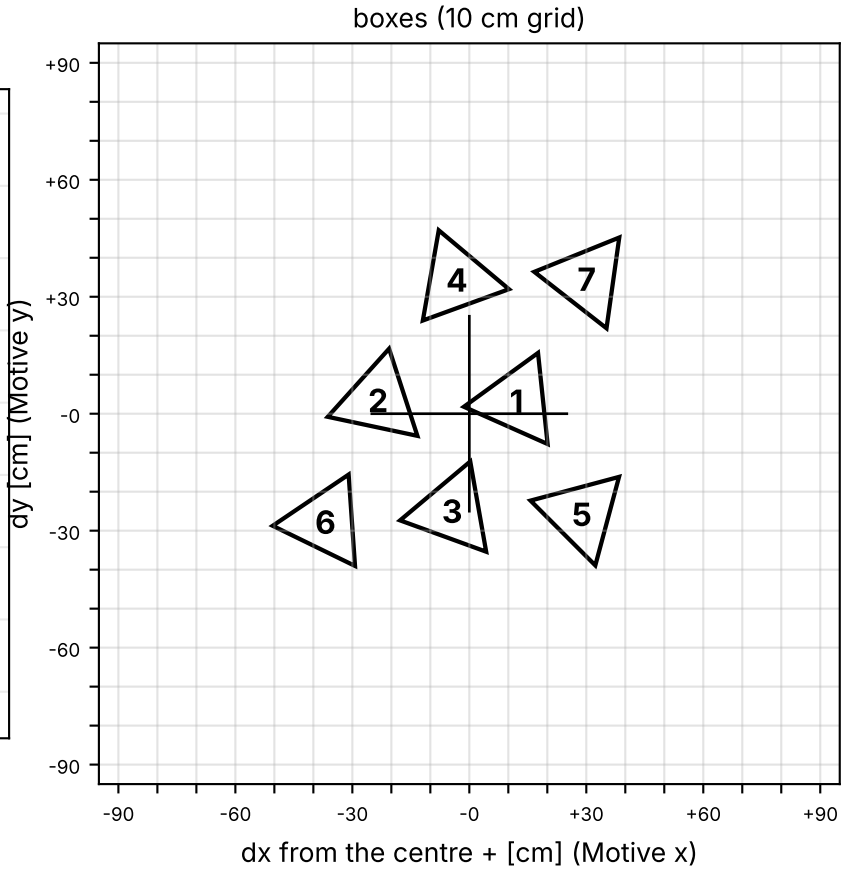
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R85 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-
avoidance --radius 1)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R86 day 6 M medium (test)

7 boxes, dense block, not touching, robots: casuale | medium in 3/4 rounds (71% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+12	+3
2	-23	+3
3	-4	-25
4	-3	+34
5	+29	-26
6	-37	-28
7	+30	+34

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	-1.21	+0.63
R2	-1.27	-0.95
R3	-0.34	-1.73
R4	+0.60	-1.62

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

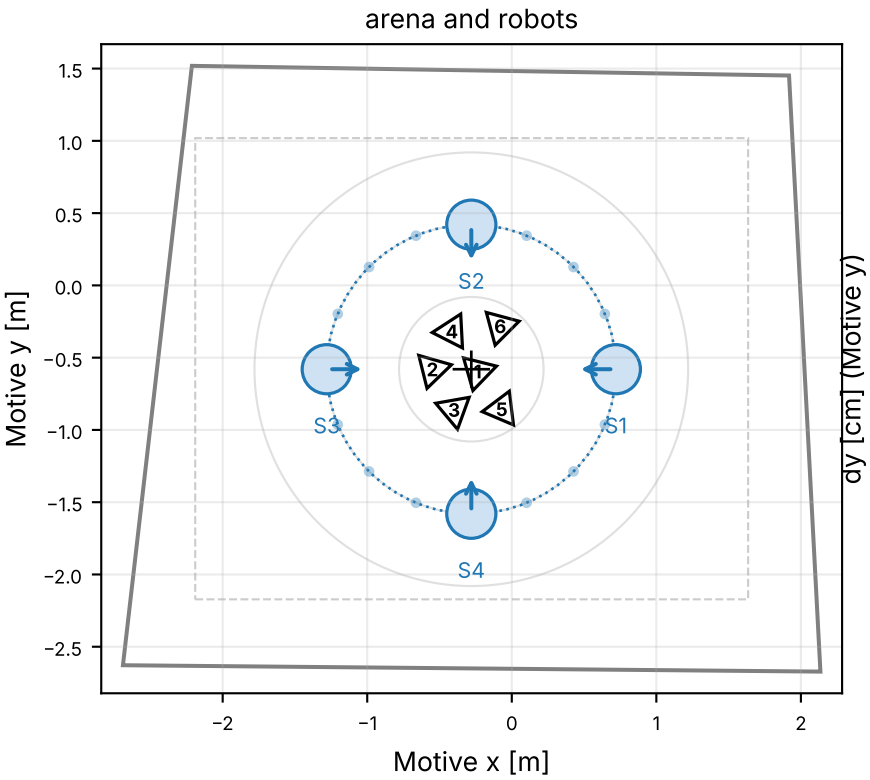
Place and capture: python -m tools.place_check --layout R86

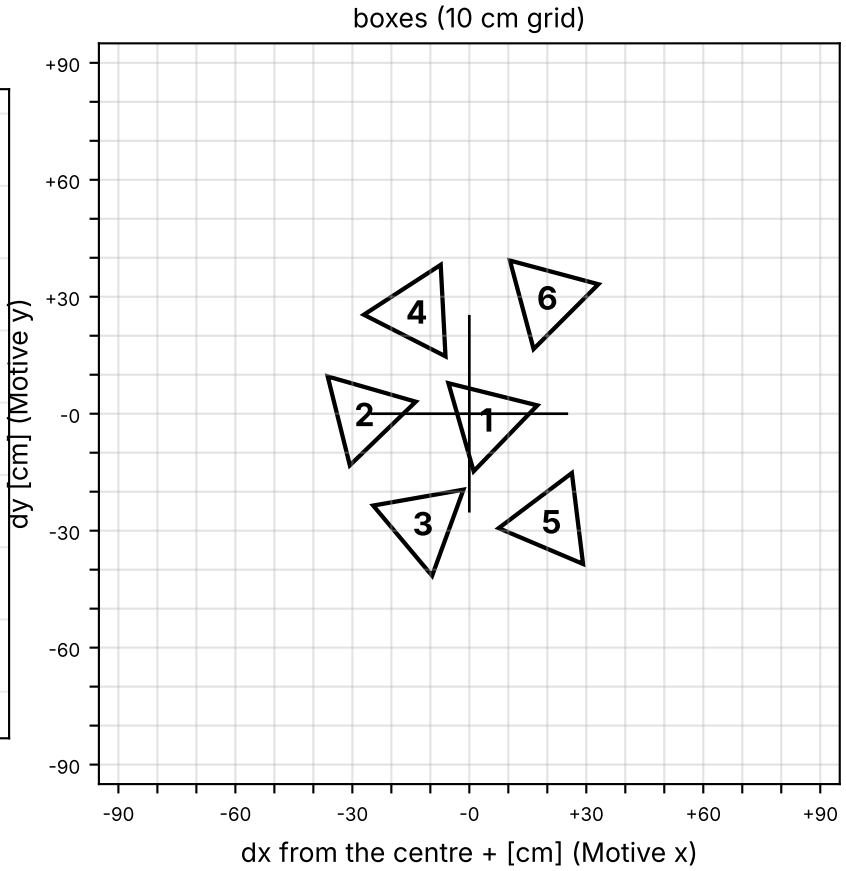
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R86 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R87 day 6 M medium (test)

6 boxes, dense block, not touching, robots: cerchio 1.0 | medium in 3/4 rounds (71% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+4	-2
2	-27	-0
3	-12	-28
4	-13	+26
5	+21	-28
6	+20	+30

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

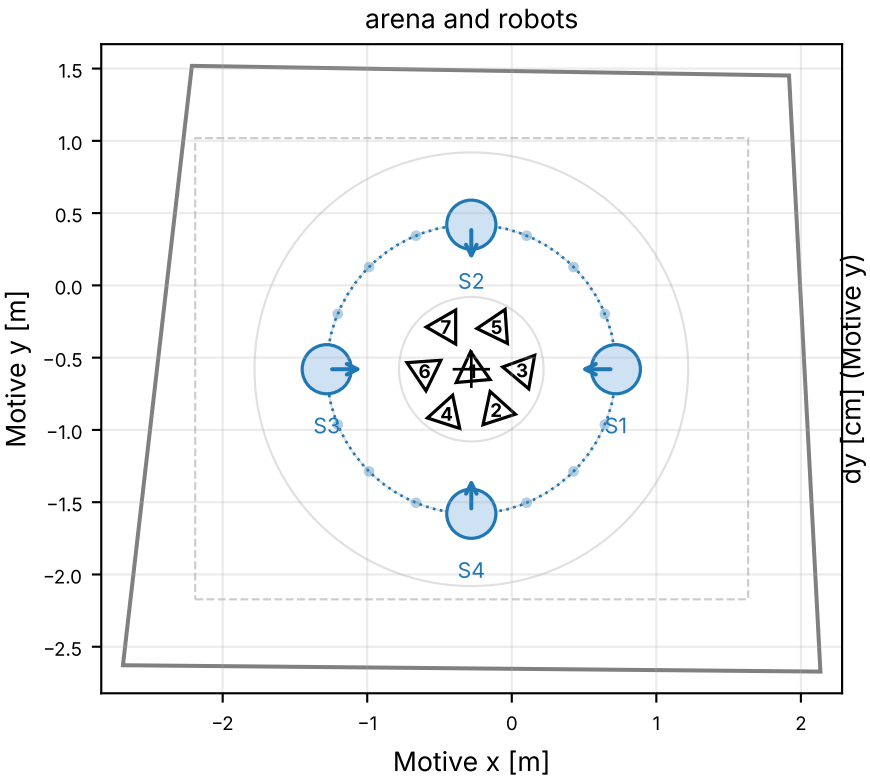
Place and capture: python -m tools.place_check --layout R87

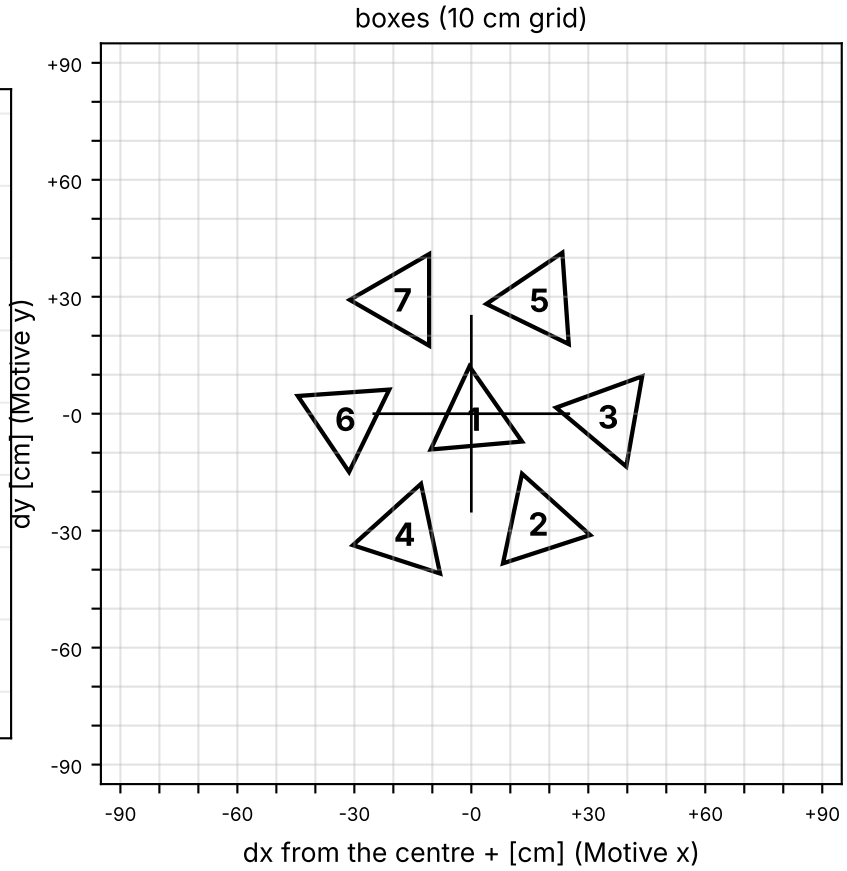
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R87 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R88 day 6 M medium (test)

7 boxes, dense block, not touching, robots: cerchio 1.0 | medium in 3/4 rounds (86% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+1	-1
2	+17	-28
3	+35	-1
4	-17	-31
5	+17	+29
6	-32	-1
7	-18	+29

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

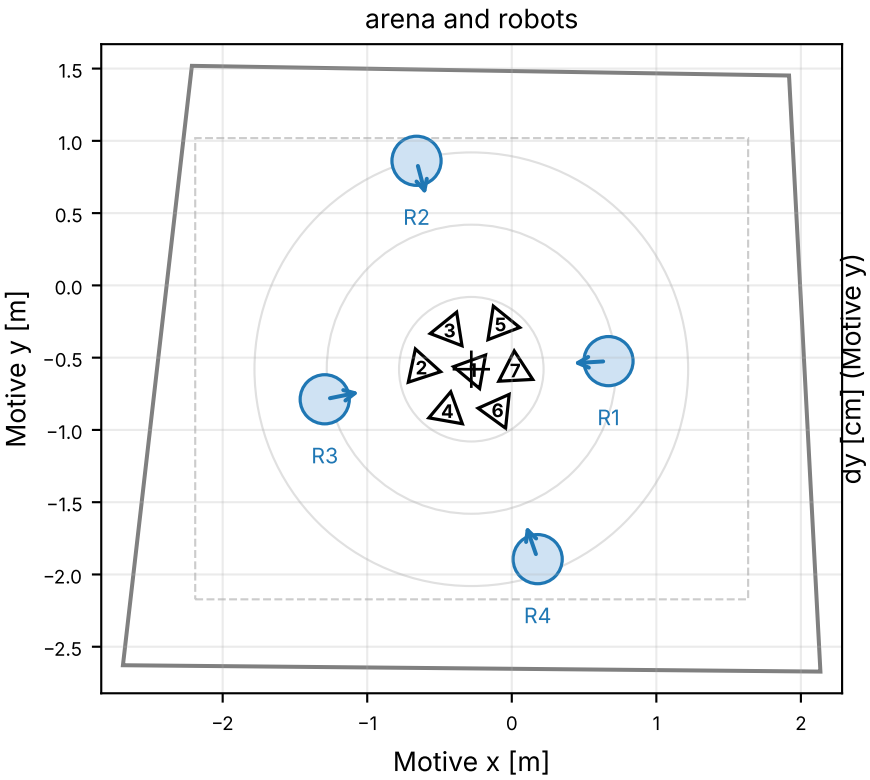
Place and capture: python -m tools.place_check --layout R88

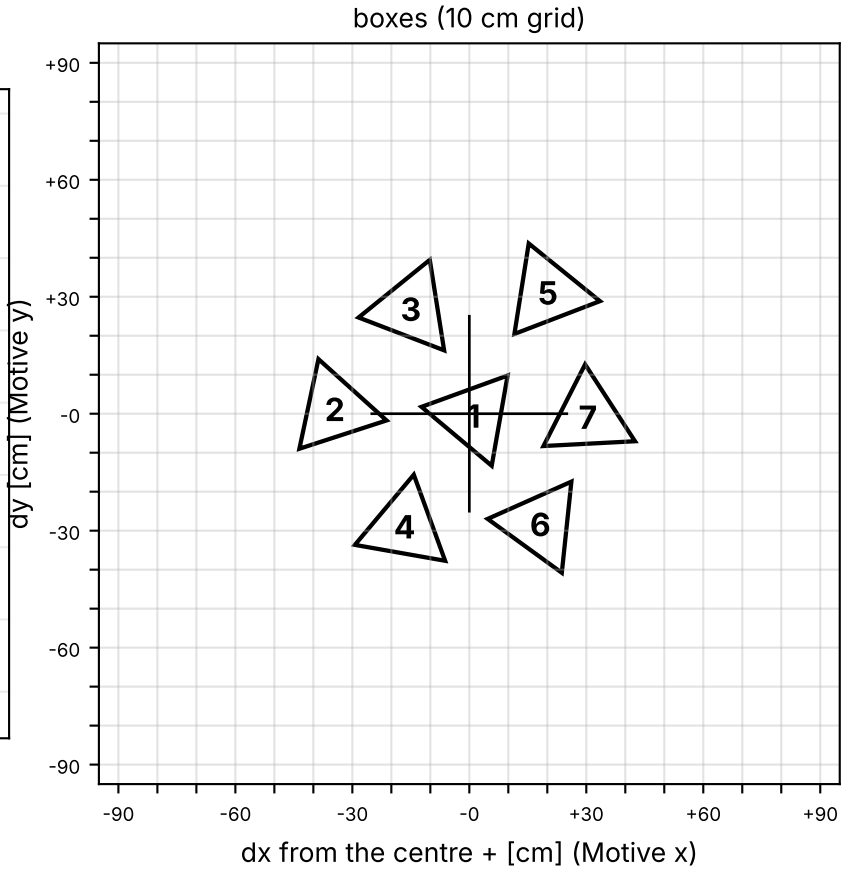
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R88 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R89 day 6 M medium (test)

7 boxes, dense block, not touching, robots: casuale | medium in 3/4 rounds (66% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+1	-1
2	-34	+1
3	-15	+27
4	-17	-29
5	+20	+31
6	+18	-28
7	+30	-1

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.67	-0.52
R2	-0.66	+0.86
R3	-1.29	-0.79
R4	+0.18	-1.89

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

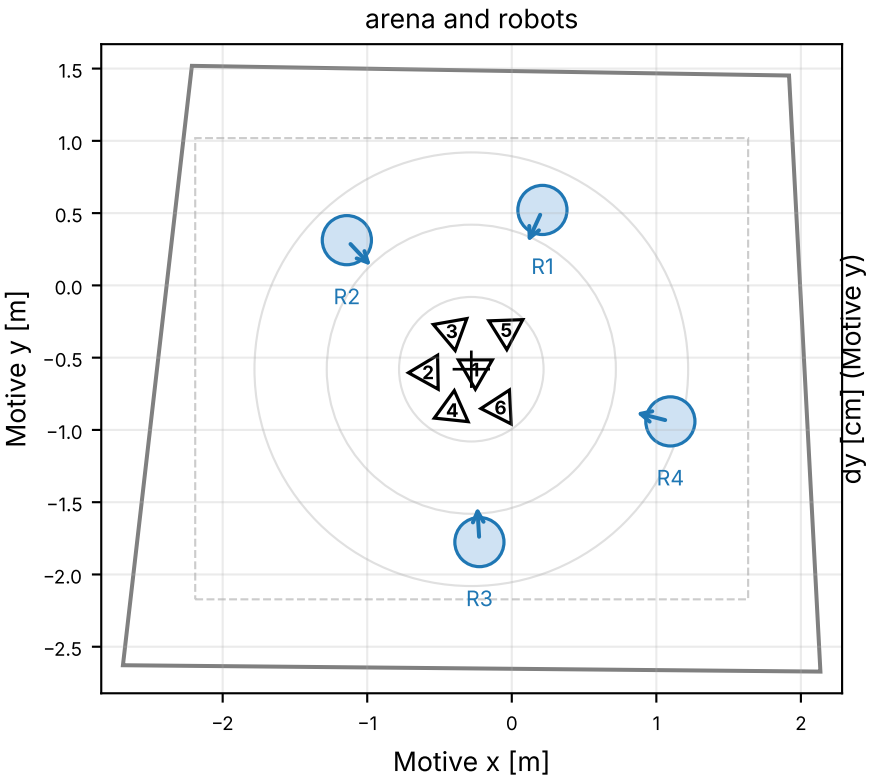
Place and capture: python -m tools.place_check --layout R89

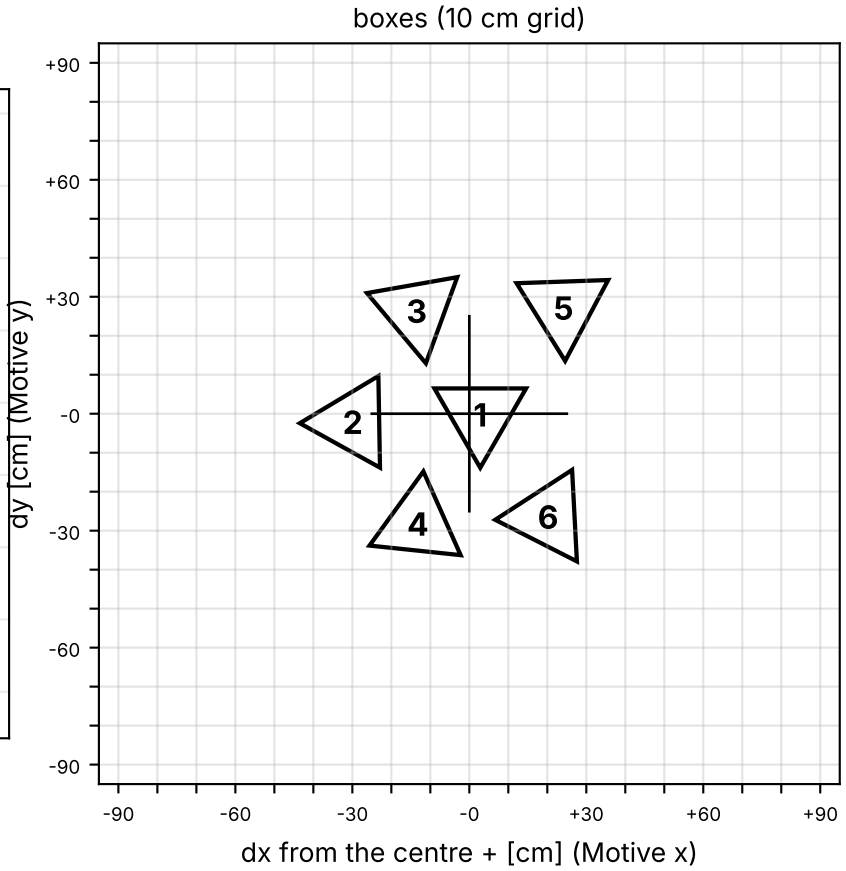
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R89 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R90 day 6 M medium (test)

6 boxes, dense block, not touching, robots: casuale | medium in 3/4 rounds (79% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+3	-0
2	-30	-2
3	-13	+26
4	-13	-28
5	+24	+27
6	+20	-26

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.21	+0.52
R2	-1.14	+0.31
R3	-0.22	-1.78
R4	+1.10	-0.94

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

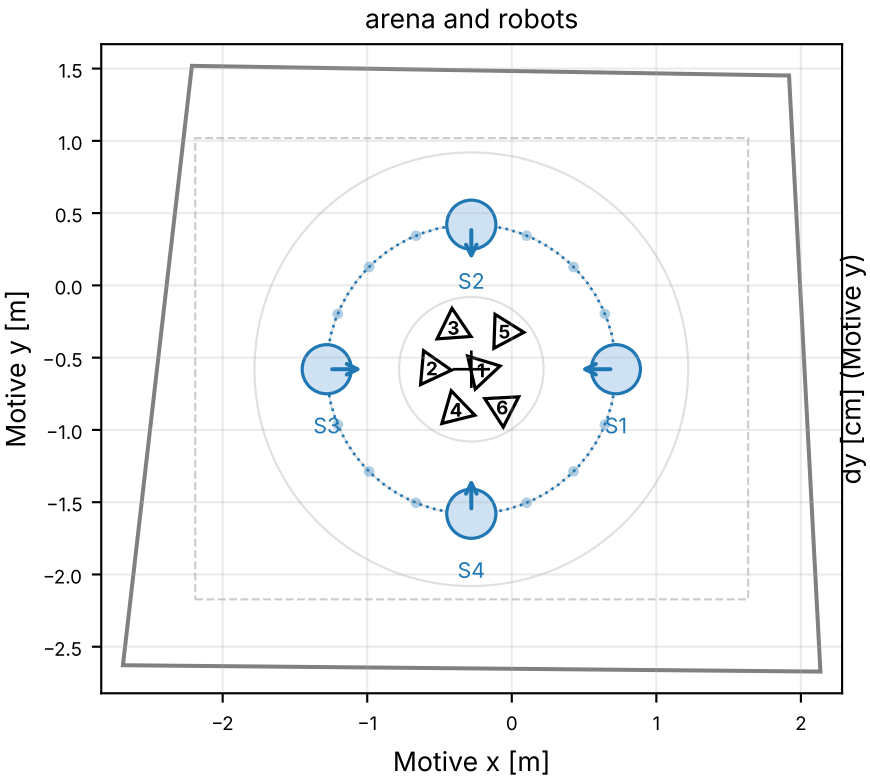
Place and capture: python -m tools.place_check --layout R90

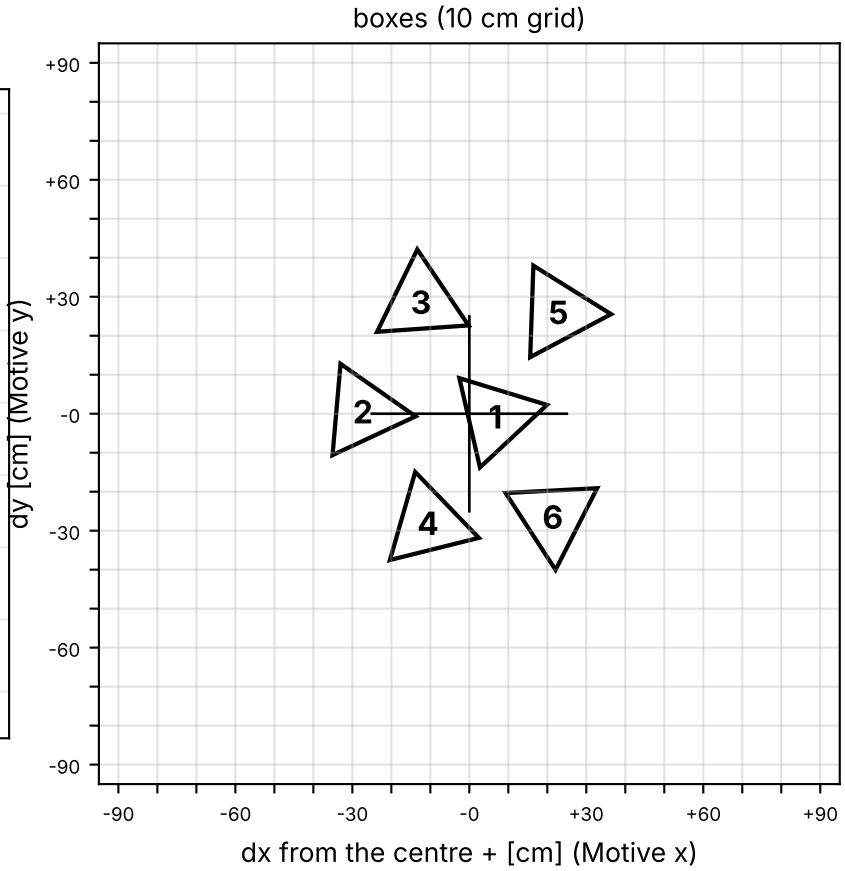
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R90 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R91 day 6 M medium (test)

6 boxes, dense block, not touching, robots: cerchio 1.0 | medium in 3/4 rounds (62% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+7	-1
2	-27	+1
3	-12	+29
4	-11	-28
5	+23	+26
6	+21	-26

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

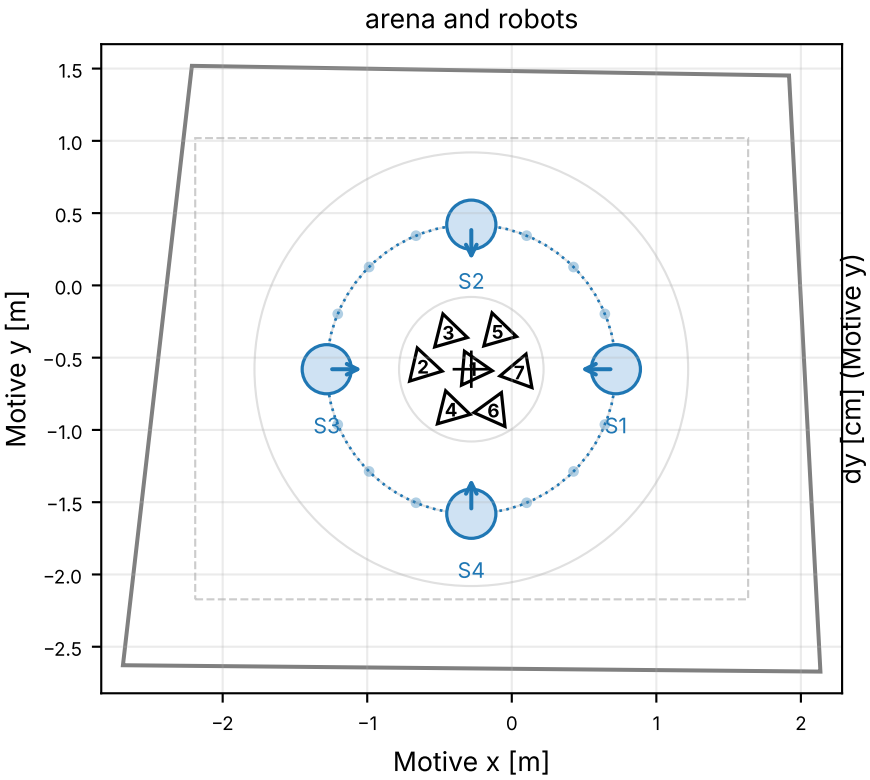
Place and capture: python -m tools.place_check --layout R91

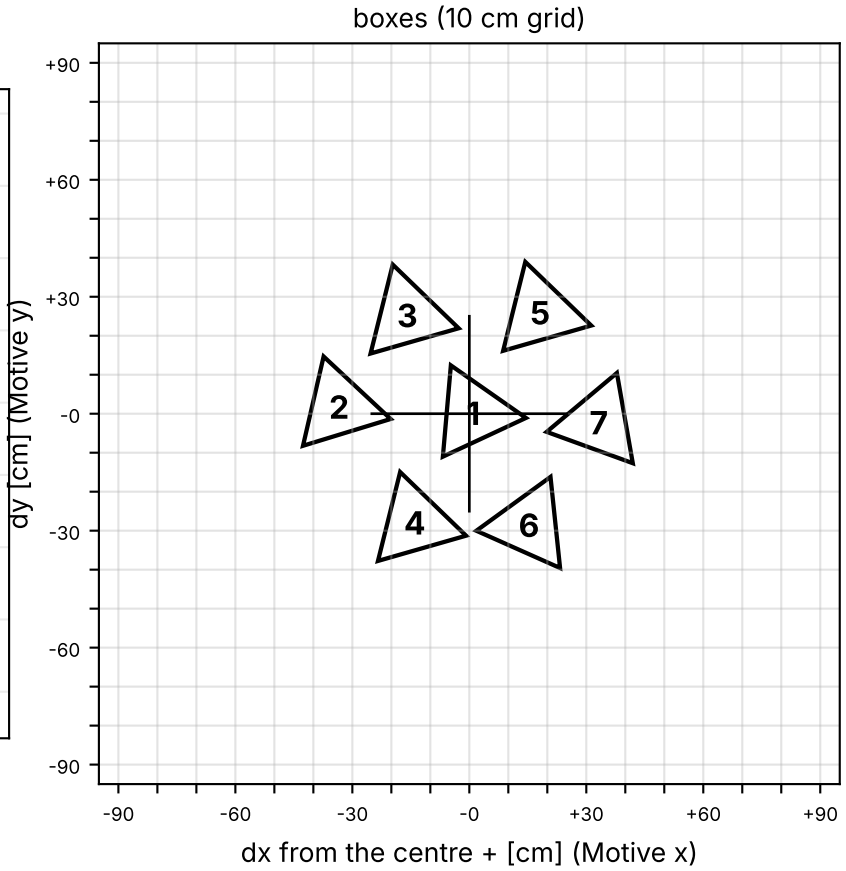
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R91 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R92 day 6 M medium (test)

7 boxes, dense block, not touching, robots: cerchio 1.0 | medium in 3/4 rounds (66% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+1	+0
2	-33	+2
3	-16	+25
4	-14	-28
5	+18	+26
6	+15	-29
7	+33	-2

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

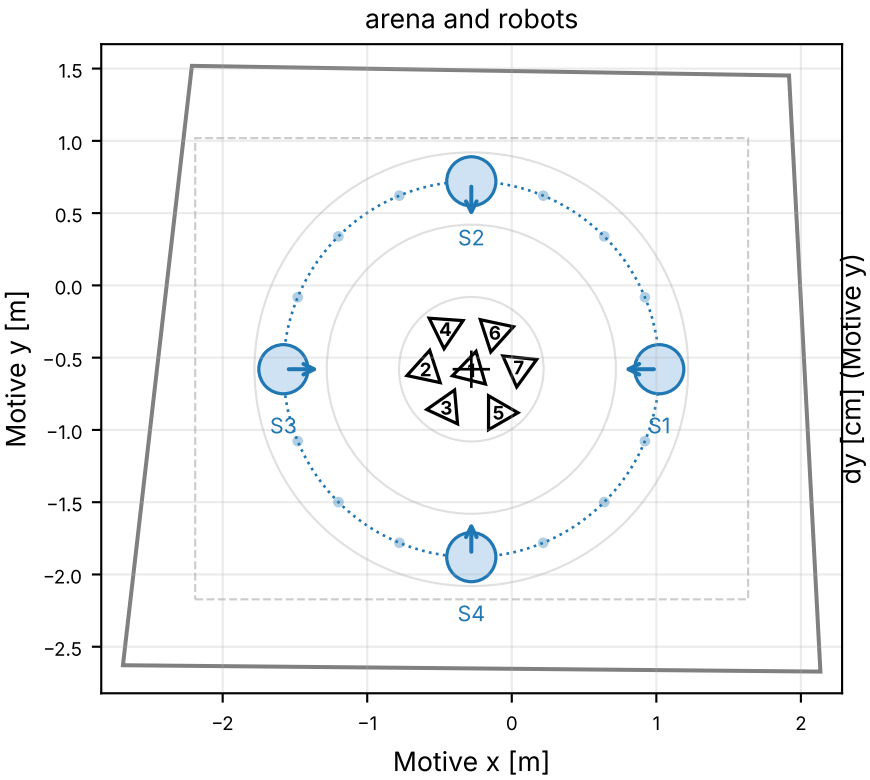
Place and capture: python -m tools.place_check --layout R92

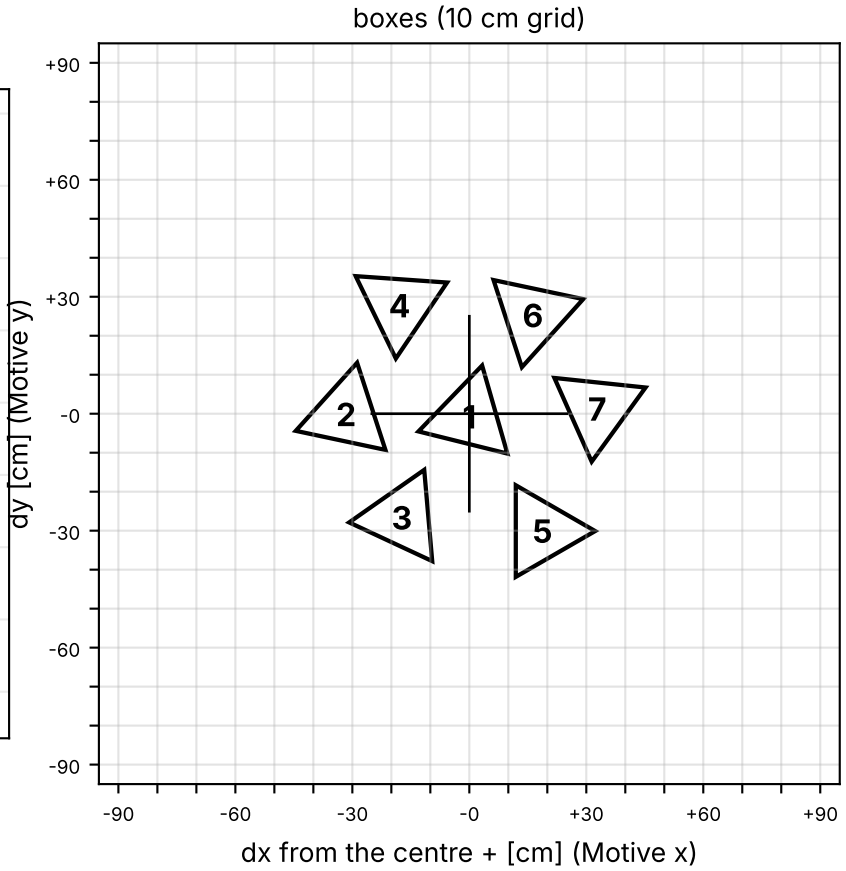
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R92 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R93 day 6 M medium (test)

7 boxes, dense block, not touching, robots: cerchio 1.3 | medium in 3/4 rounds (62% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+0	-1
2	-32	-0
3	-17	-27
4	-18	+28
5	+19	-30
6	+16	+25
7	+33	+1

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

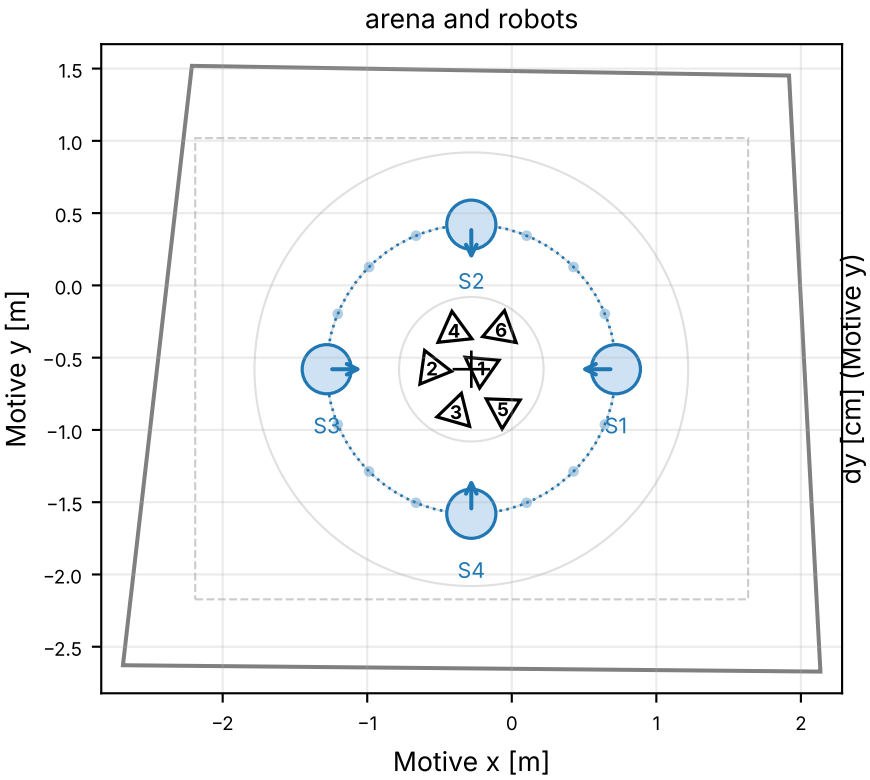
Place and capture: python -m tools.place_check --layout R93

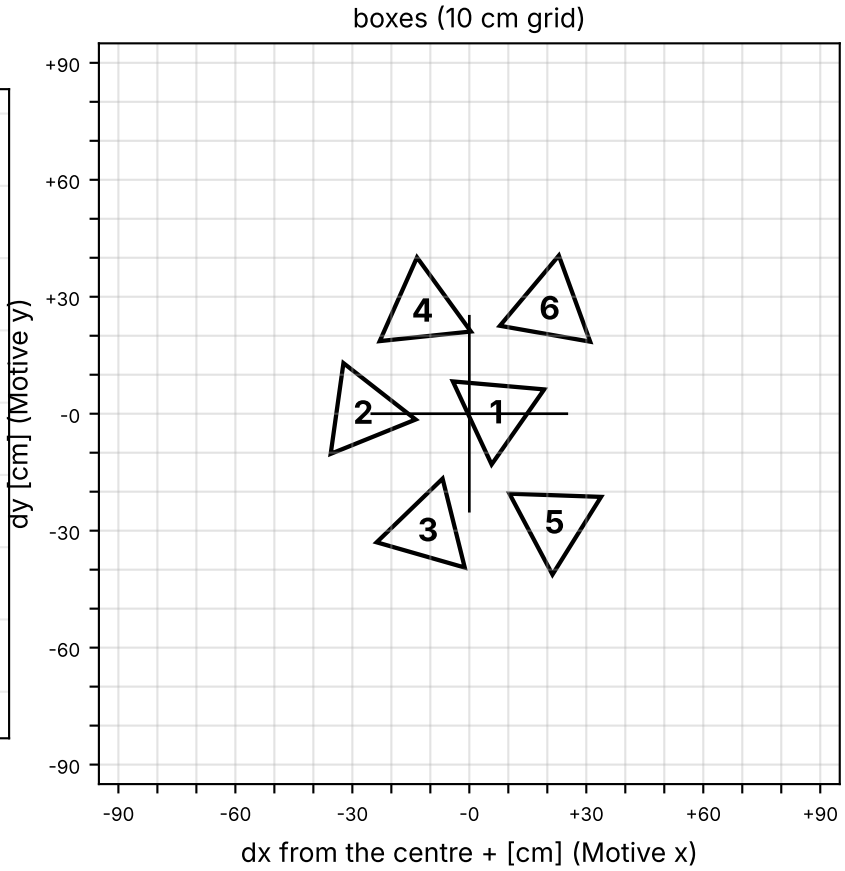
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R93 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-
avoidance --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R94 day 6 M medium (test)

6 boxes, dense block, not touching, robots: cerchio 1.0 | medium in 3/4 rounds (66% of the draws)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+7	+1
2	-27	+0
3	-11	-30
4	-12	+27
5	+22	-28
6	+21	+27

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

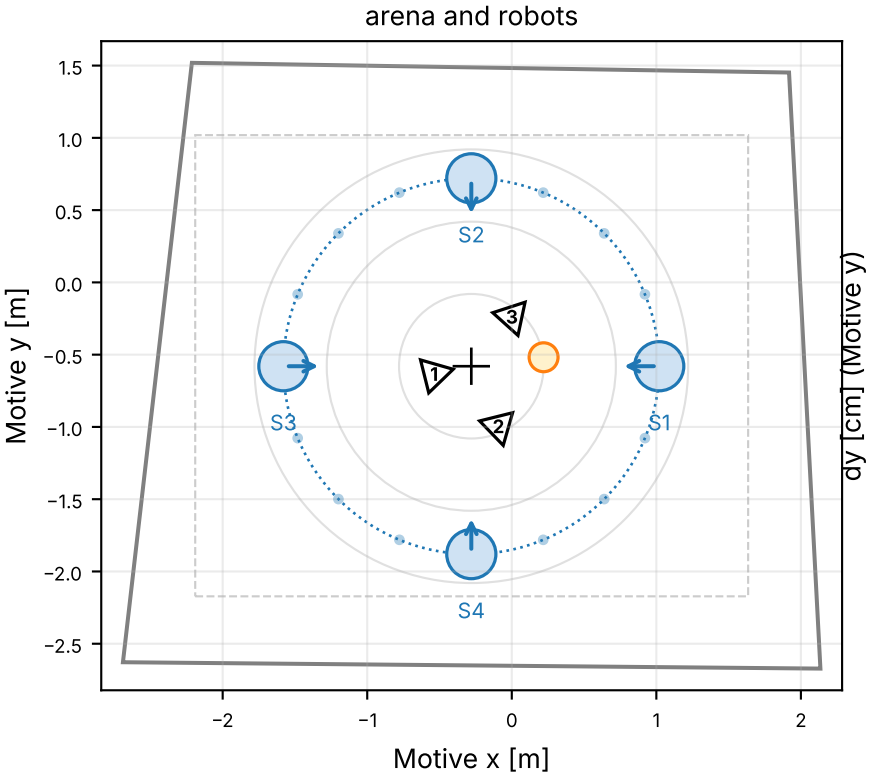
Place and capture: python -m tools.place_check --layout R94

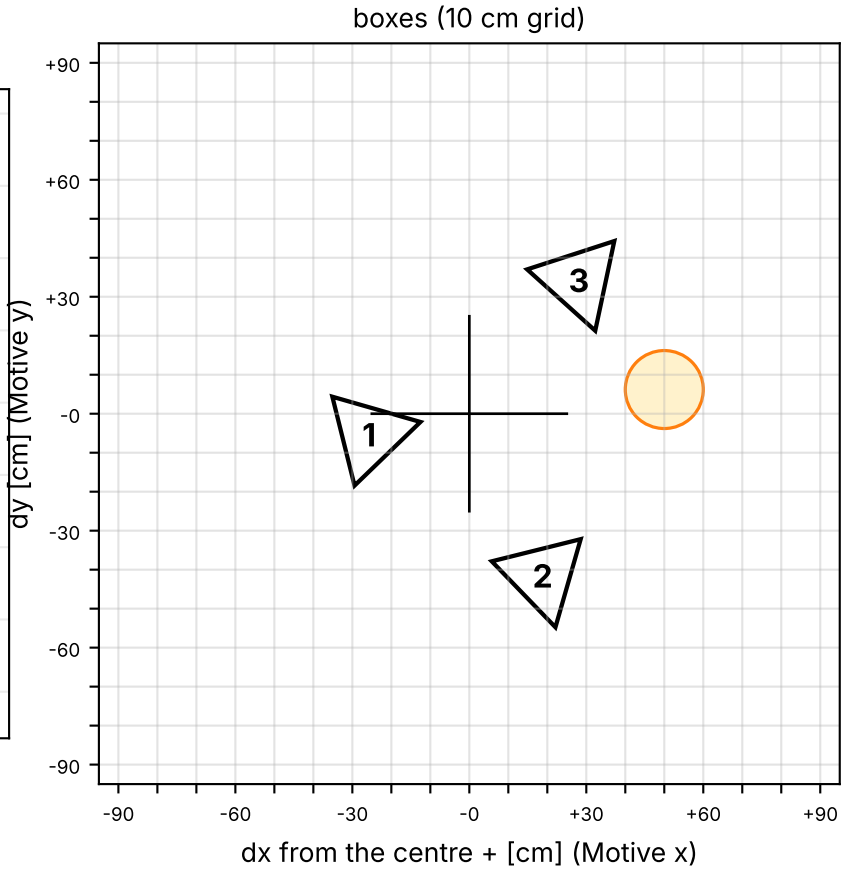
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R94 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R95 day 6 D white objects (test)

3 boxes, sparse, robots: cerchio 1.3 | 1 white object(s) that are not boxes (paper bin, white bag, white cup...)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-26	-5
2	+19	-42
3	+28	+34

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Other objects

white object	+50	+6

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

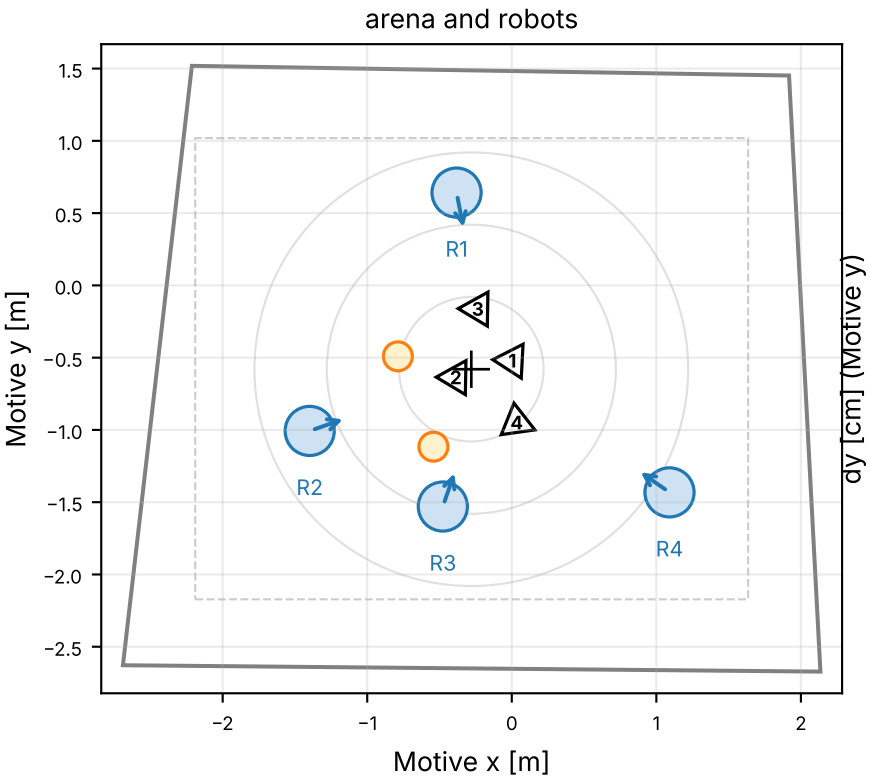
Place and capture: python -m tools.place_check --layout R95

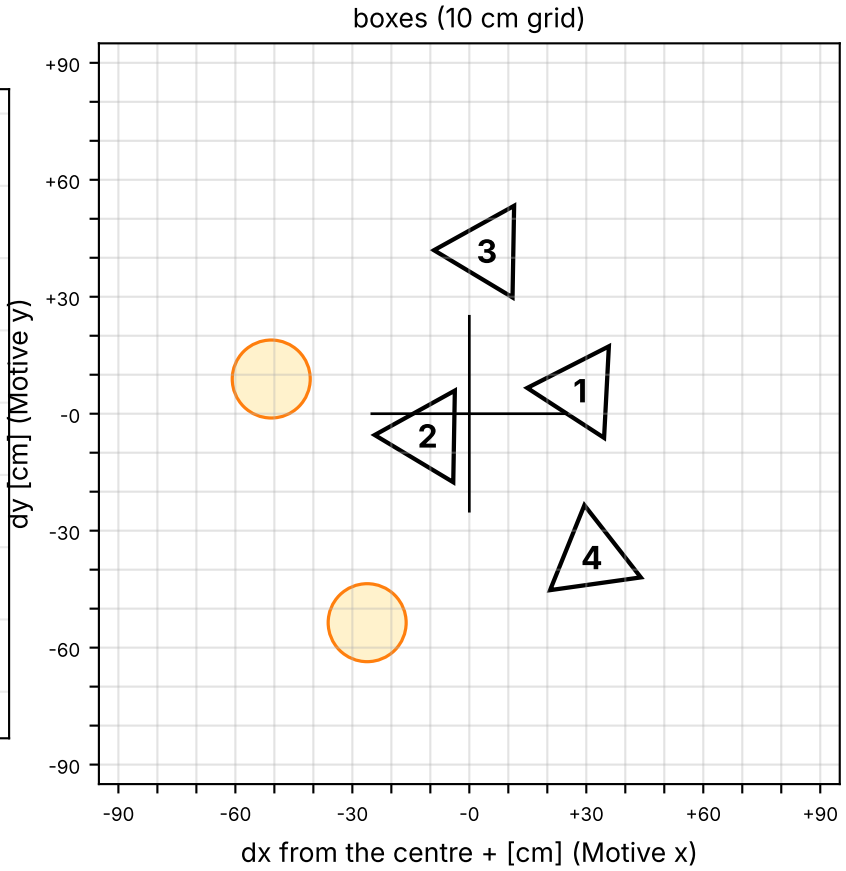
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R95 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R96 day 6 D white objects (test)

4 boxes, sparse, robots: casuale | 2 white object(s) that are not boxes (paper bin, white bag, white cup...)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+28	+6
2	-11	-6
3	+5	+42
4	+31	-37

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	-0.38	+0.64
R2	-1.40	-1.01
R3	-0.48	-1.53
R4	+1.09	-1.43

Other objects

white object	-26	-54
white object	-51	+9

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

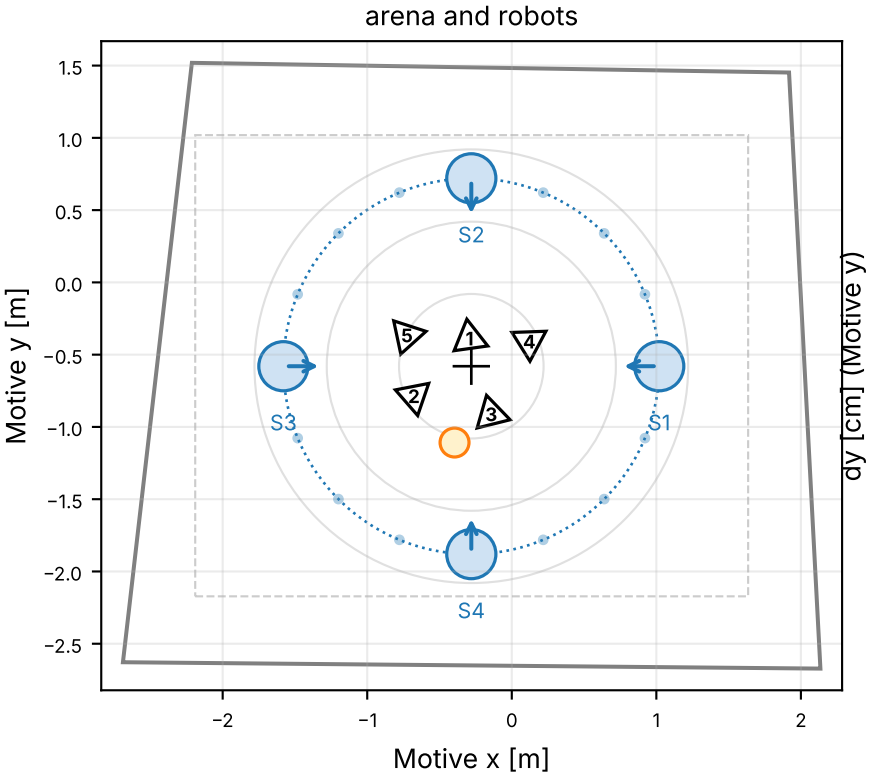
Place and capture: python -m tools.place_check --layout R96

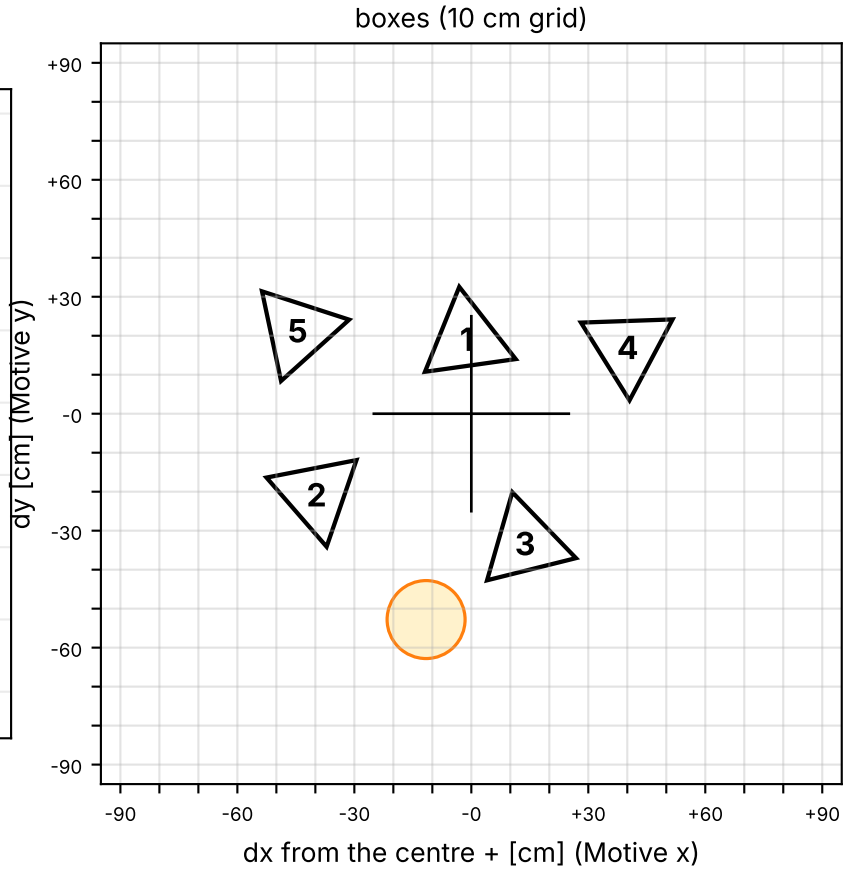
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R96 --box 11:box --box 12:box --box 13:box --box 14:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R97 day 6 D white objects (test)

5 boxes, sparse, robots: cerchio 1.3 | 1 white object(s) that are not boxes (paper bin, white bag, white cup...)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-1	+19
2	-40	-21
3	+14	-33
4	+40	+17
5	-45	+21

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Other objects

white object	-12	-53

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

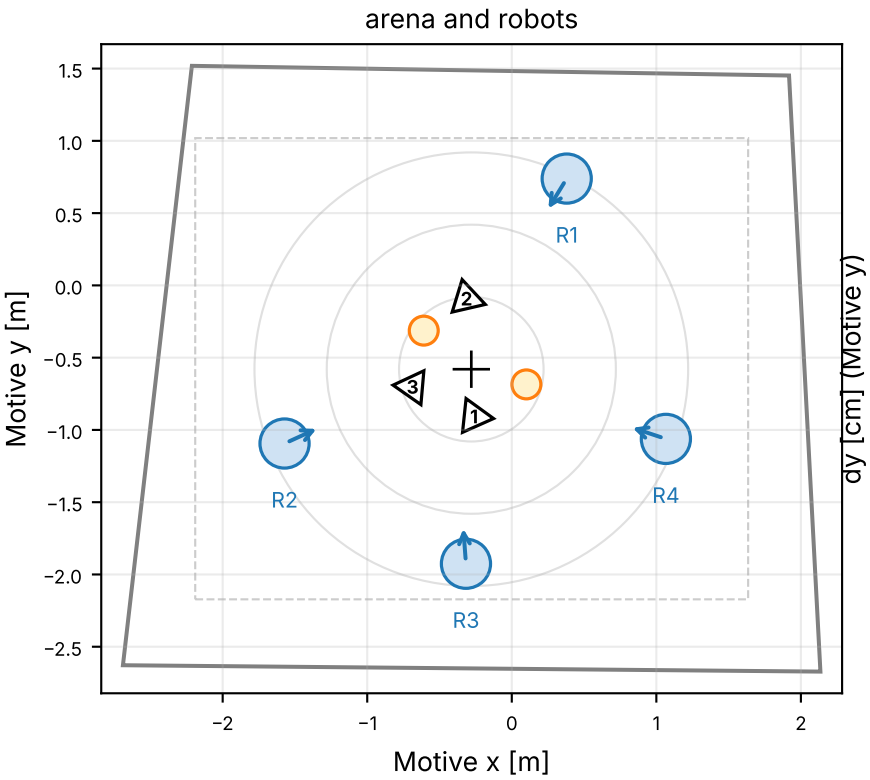
Place and capture: python -m tools.place_check --layout R97

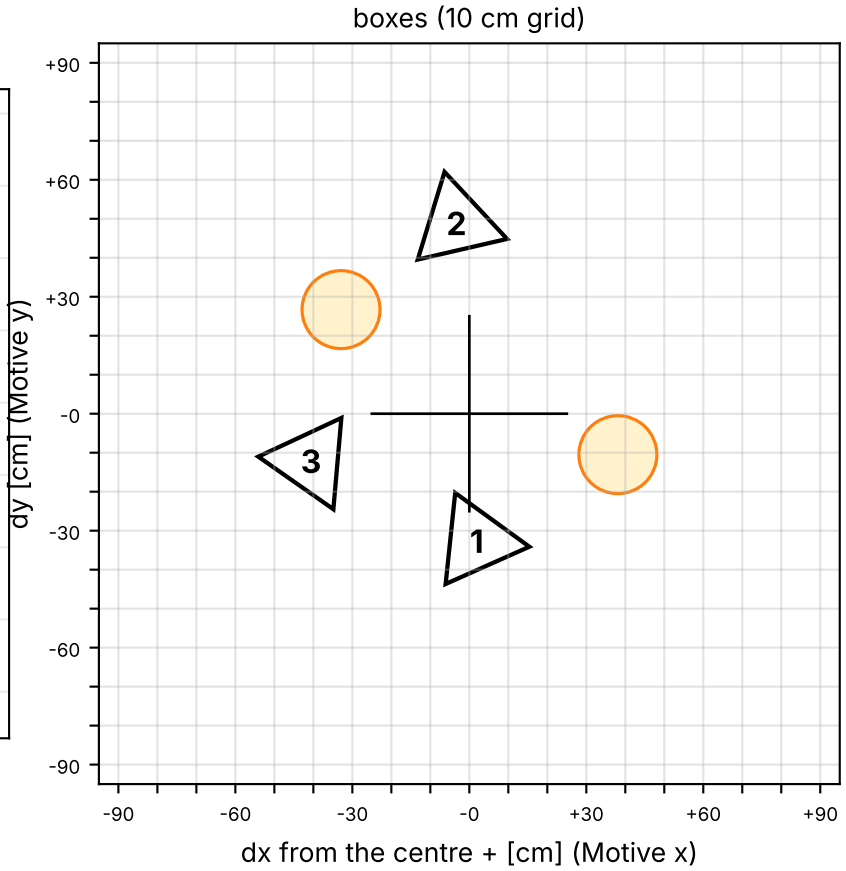
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R97 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R98 day 6 D white objects (test)

3 boxes, sparse, robots: casuale | 2 white object(s) that are not boxes (paper bin, white bag, white cup...)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+2	-33
2	-3	+49
3	-41	-12

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.38	+0.74
R2	-1.57	-1.09
R3	-0.32	-1.93
R4	+1.07	-1.06

Other objects

white object	+38	-11
white object	-33	+27

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

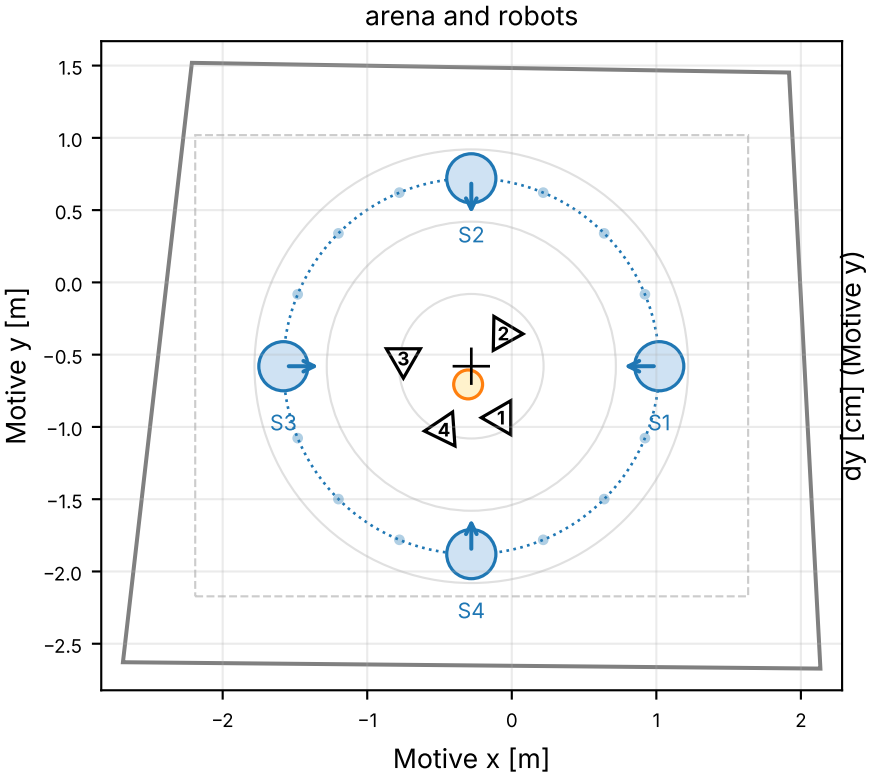
Place and capture: python -m tools.place_check --layout R98

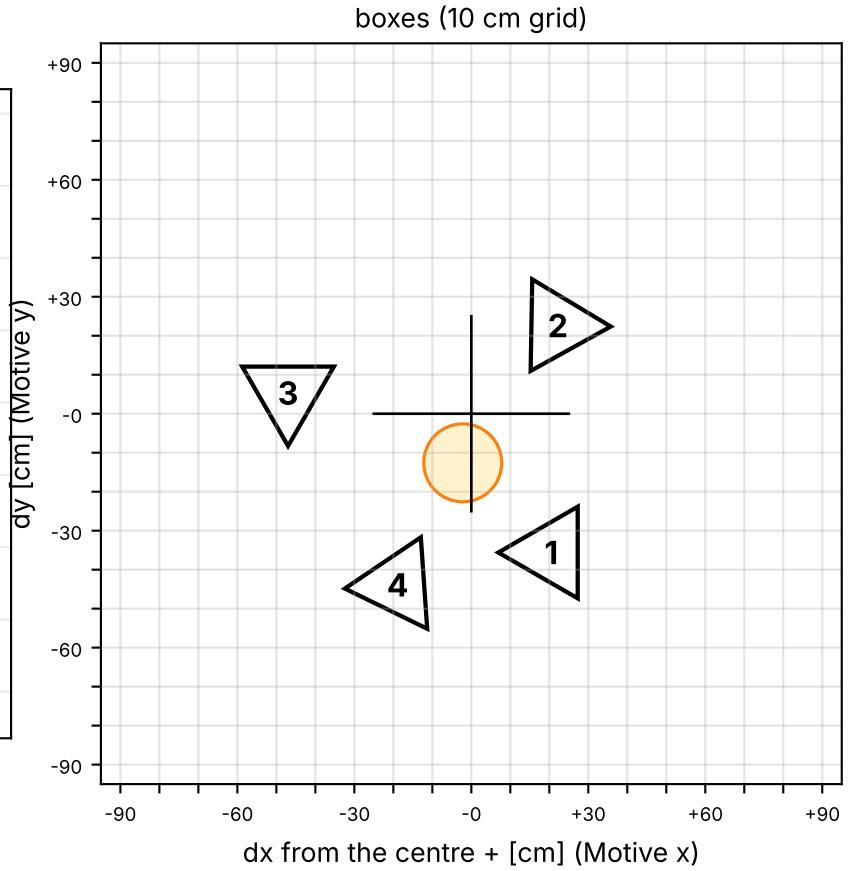
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R98 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R99 day 6 D white objects (test)

4 boxes, sparse, robots: cerchio 1.3 | 1 white object(s) that are not boxes (paper bin, white bag, white cup...)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+20	-36
2	+22	+23
3	-47	+5
4	-19	-44

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Other objects

white object	-2	-13

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

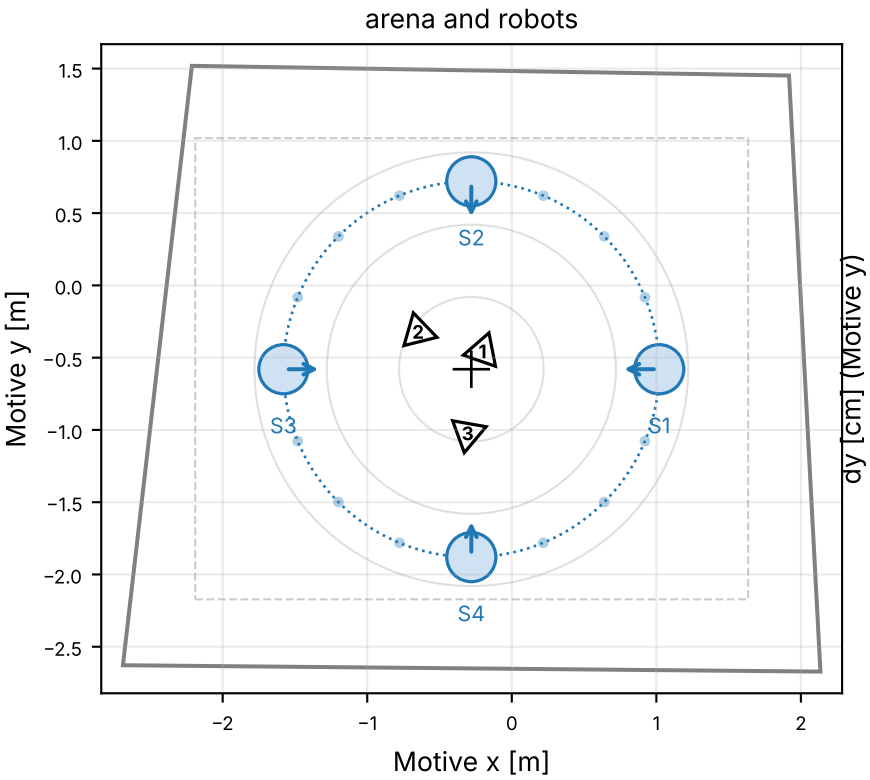
Place and capture: python -m tools.place_check --layout R99

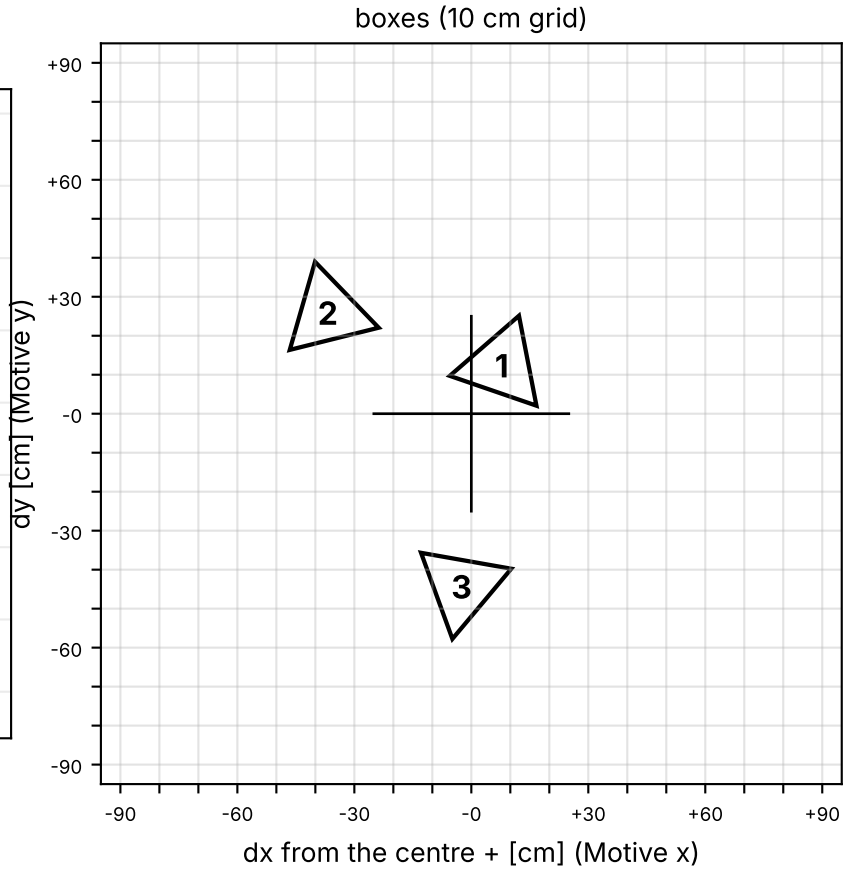
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R99 --box 11:box --box 12:box --box 13:box --box 14:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R100 day 6 E light (test)

3 boxes, sparse, robots: cerchio 1.3 | curtains open





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+8	+12
2	-37	+26
3	-2	-44

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

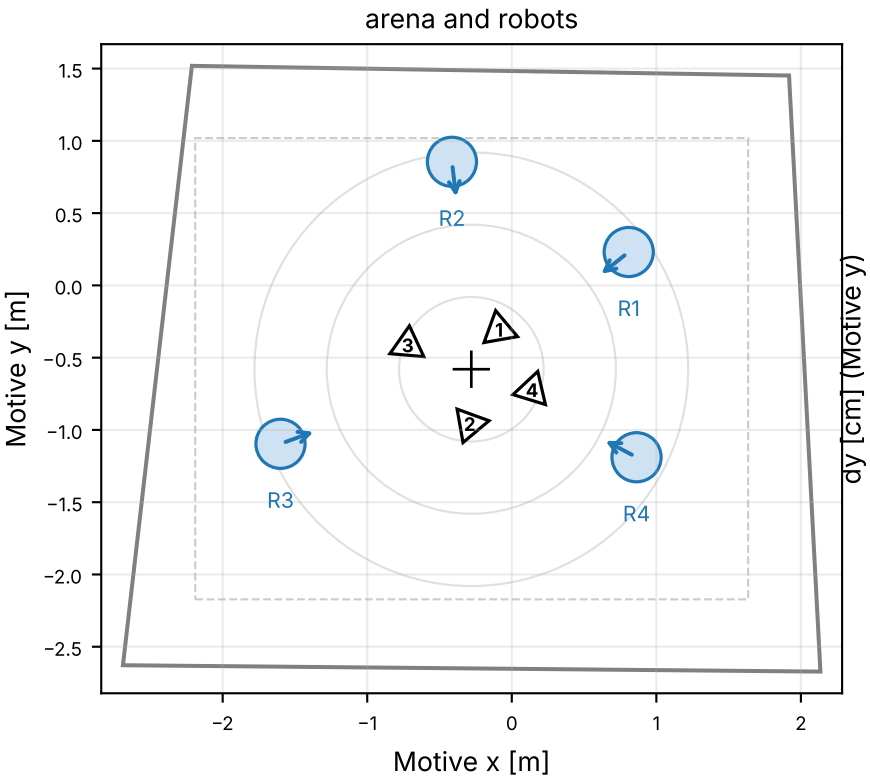
Place and capture: python -m tools.place_check --layout R100

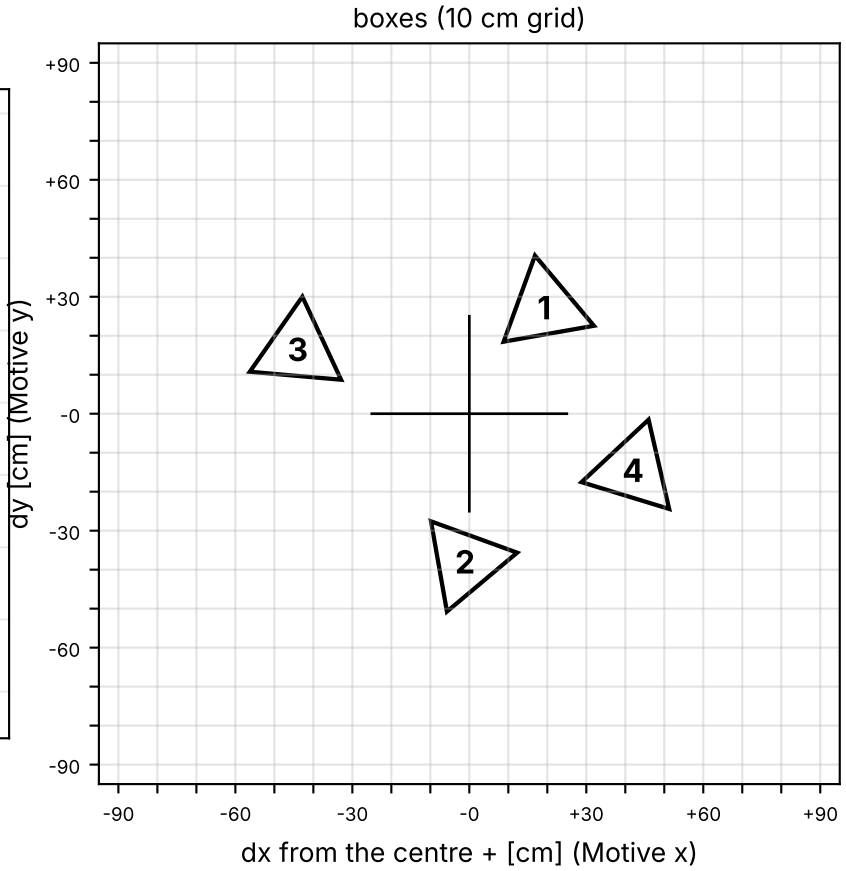
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R100 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R101 day 6 E light (test)

4 boxes, sparse, robots: casuale | half of the lights off





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+19	+27
2	-1	-38
3	-44	+16
4	+42	-15

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.81	+0.23
R2	-0.41	+0.85
R3	-1.60	-1.10
R4	+0.86	-1.19

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

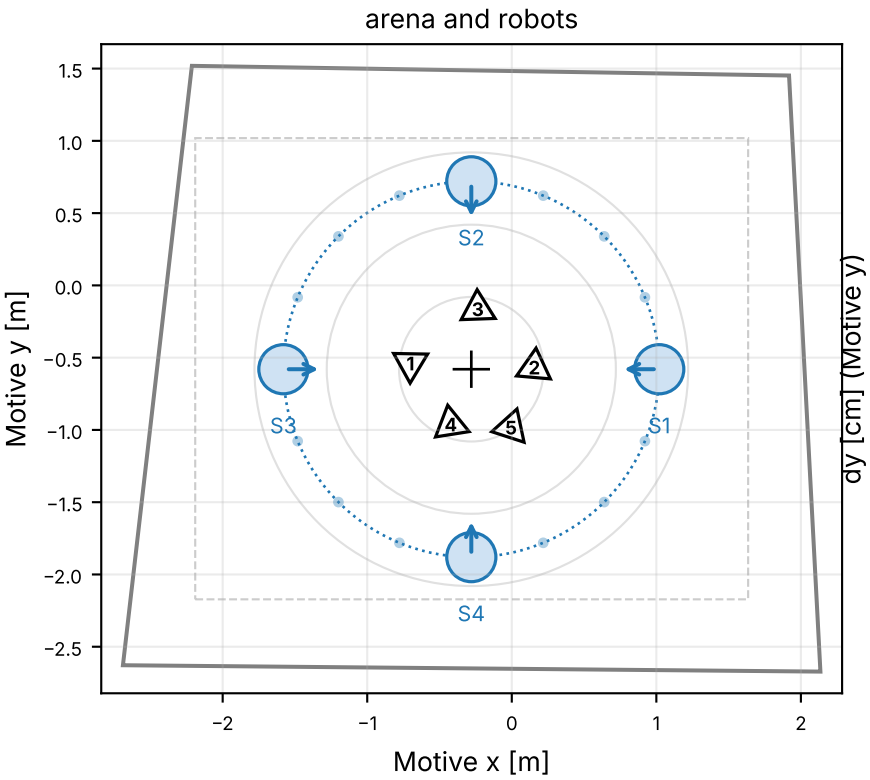
Place and capture: python -m tools.place_check --layout R101

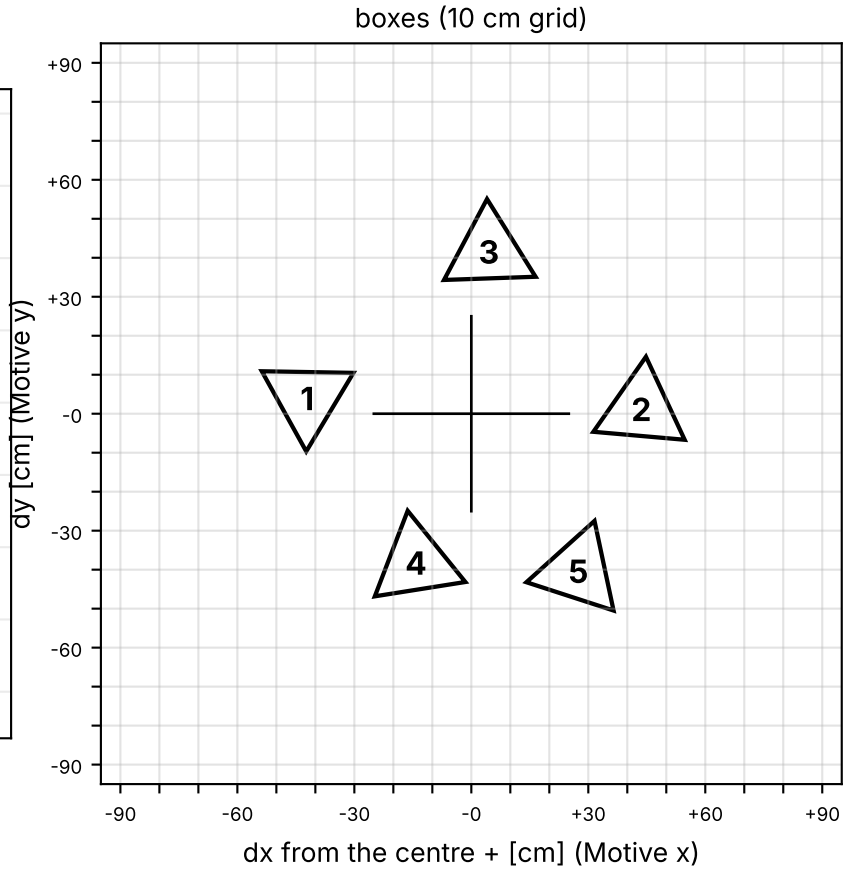
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R101 --box 11:box --box 12:box --box 13:box --box 14:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R102 day 6 E light (test)

5 boxes, sparse, robots: cerchio 1.3 | curtains open





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-42	+4
2	+44	+1
3	+5	+41
4	-14	-38
5	+27	-40

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

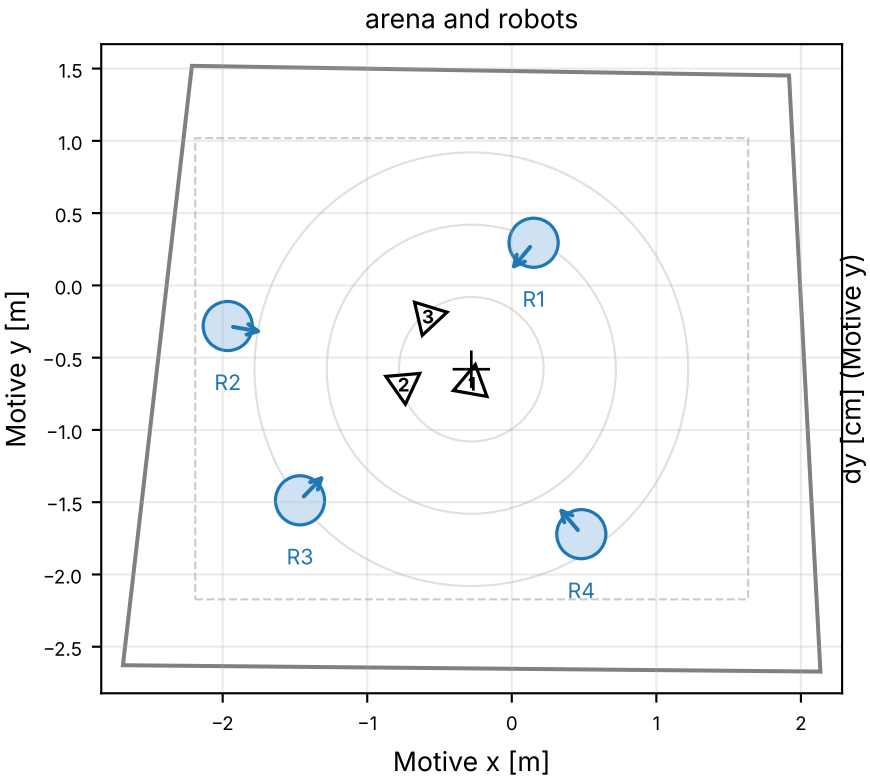
Place and capture: python -m tools.place_check --layout R102

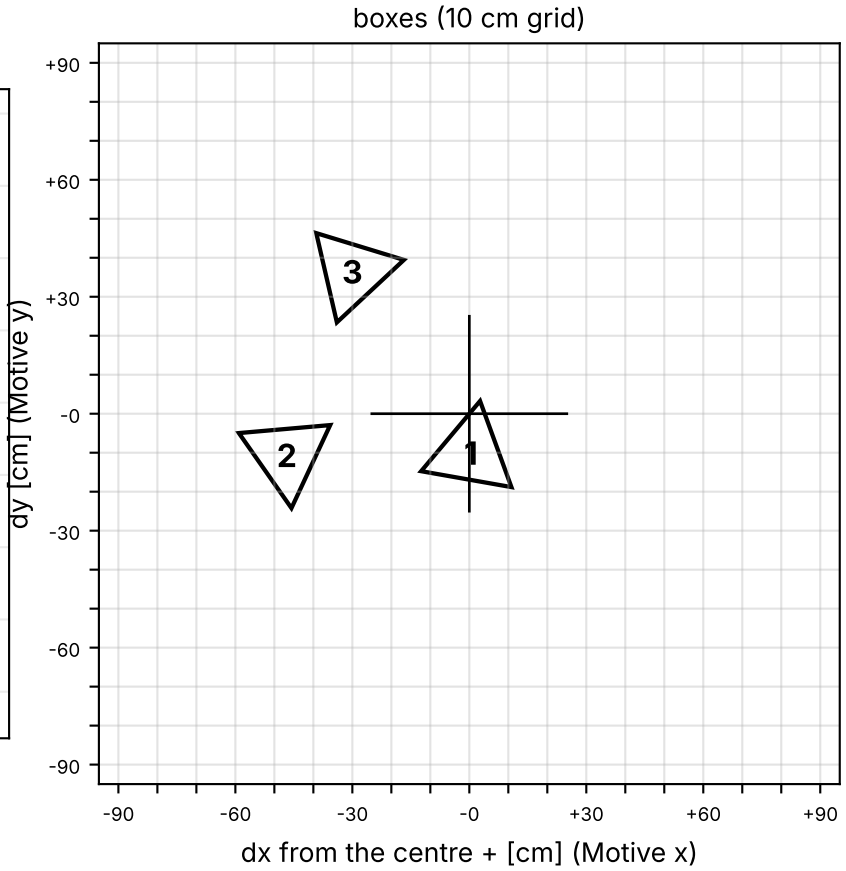
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R102 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R103 day 6 E light (test)

3 boxes, sparse, robots: casuale | half of the lights off





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+0	-10
2	-47	-11
3	-30	+36

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.15	+0.29
R2	-1.97	-0.28
R3	-1.47	-1.49
R4	+0.48	-1.72

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

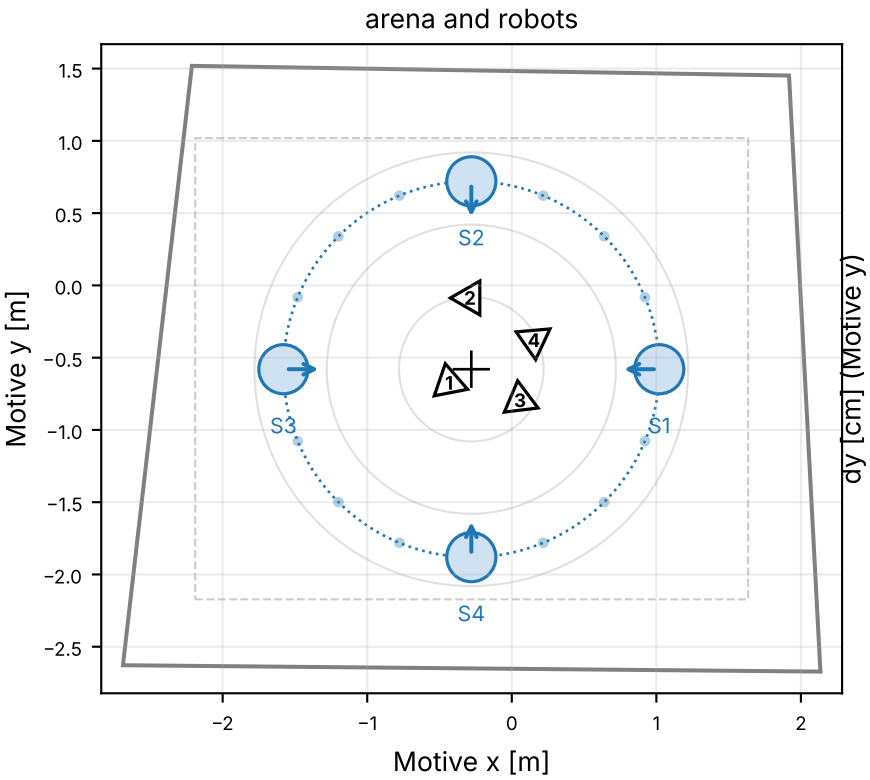
Place and capture: `python -m tools.place_check --layout R103`

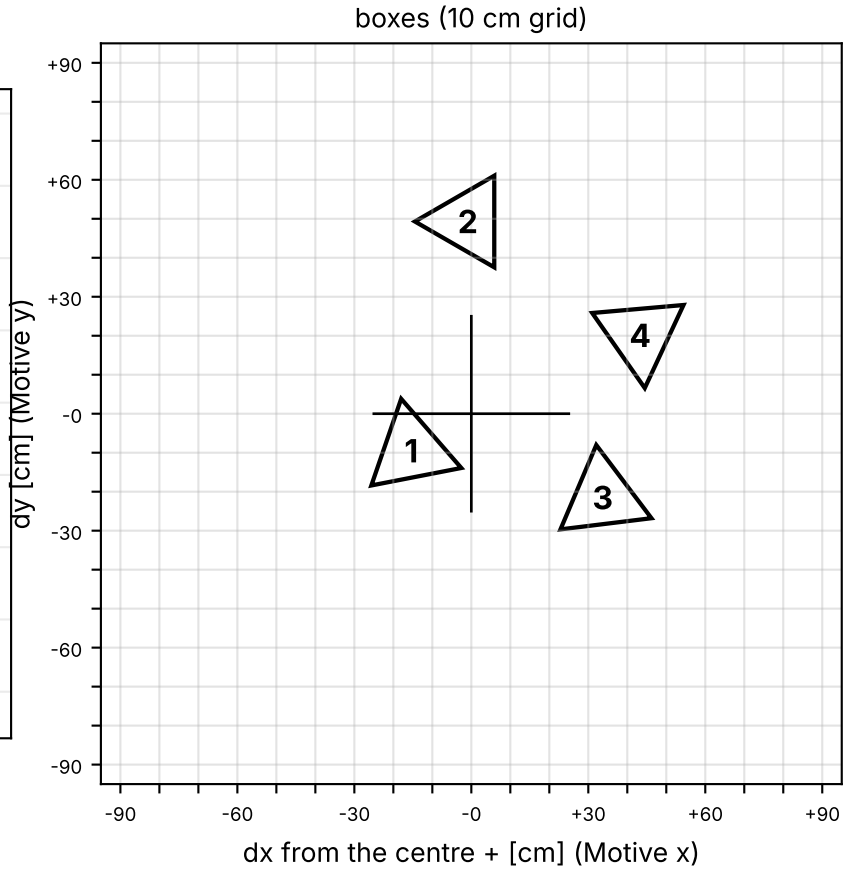
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R103 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R104 day 6 E light (test)

4 boxes, sparse, robots: cerchio 1.3 | curtains open





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-15	-10
2	-1	+49
3	+34	-22
4	+43	+20

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

Place and capture: python -m tools.place_check --layout R104

(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R104 --box 11:box --box 12:box --box 13:box --box 14:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --circle-center -0.28 -0.58 --no-robot-avoidance --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV